TM Forum Specification

Resource Activation and Configuration User Guide

TMF702

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Notice

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Table of Contents

Notice . . . i
Introduction . . . 1
Scope . . . 1
Relationship to Resource Inventory (TMF639) . . . 1
Operations . . . 1
Sample Use Cases . . . 3
Support of polymorphism and extension patterns . . . 5
RESOURCE MODEL . . . 6
Managed Entity and Task Resource Models . . . 6
Resource resource . . . 6
Monitor resource . . . 58
PhysicalResource resource . . . 62
LogicalResource resource . . . 90
Notification Resource Models . . . 108
Resource . . . 110
Monitor . . . 116
PhysicalResource . . . 118
LogicalResource . . . 123
API OPERATIONS . . . 130
Operations on Resource . . . 130
Retrieves a resource by ID . . . 130
List or find resource objects . . . 132
Creates a resource . . . 133
Updates partially a resource . . . 135
Deletes a resource . . . 140
Operations on Monitor . . . 141
Retrieves a monitor by ID . . . 141
List or find monitor objects . . . 142
API NOTIFICATIONS . . . 145
Register listener . . . 145
Unregister listener . . . 146
Publish Event to listener . . . 146
Acknowledgements . . . 148

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Release History 148
Contributors to Document 148

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Introduction

The following document is intended to provide details of the REST API interface for the Resource Activation and Configuration Management. The intent of this API is to provide a consistent and standardized mechanism to activate, configure, and manage the state of Resources in the network.

It includes the model definition as well as all available operations.

Scope

The Resource Activation and Configuration API (TMF702) provides a means for a system to activate and configure network resources, and to track the status of those operations. The API distinguishes between two categories of resource:

- **PhysicalResource** – tangible hardware assets such as routers, switches, antennas, and cables.
- **LogicalResource** – software-defined or virtual resources such as IP addresses, MSISDN numbers, VLANs, and software licenses.

The API also provides a **Monitor** resource that enables asynchronous tracking of long-running activation and configuration operations.

Relationship to Resource Inventory (TMF639)

The Resource Activation and Configuration API complements the Resource Inventory API (TMF639). While TMF639 provides a read-oriented view of the current state of resources in the inventory, TMF702 is focused on the lifecycle management of those resources — specifically their activation, configuration changes, and decommissioning. Resources activated through TMF702 are typically reflected in the inventory managed by TMF639.

NOTE	The API model for the Resource entity used in TMF702 is essentially the same as that used in TMF639 Resource Inventory.

Operations

Resource Activation and Configuration API performs the following operations on the resources :

- Retrieve an entity or a collection of entities depending on filter criteria
- Partial update of an entity (including updating rules)

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- Create an entity (including default values and creation rules)
- Delete an entity (for administration purposes)
- Manage notification of events

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Sample Use Cases

Reader will find example of use cases using the Resource Activation and Configuration API in the "How to use ODA – Using Open APIs to realize Use Cases v29.0.0 (IG1228)" document.

Activate a subscriber's mobile number (Use Case 1)

When a customer subscribes to a mobile service, the operator's Order Management system invokes the Resource Activation and Configuration API to activate the MSISDN (a LogicalResource) on the network. The API call creates or updates the resource record, sets its `administrativeState` to `unlocked`, `operationalState` to `enabled`, and `usageState` to `active`. A `ResourceCreateEvent` or `ResourceAttributeValueChangeEvent` notification is published to downstream systems.

Commission and configure new network equipment (Use Case 2)

After physical installation of new network equipment (a PhysicalResource such as a router or base station), the Network Operations System calls the Resource Activation and Configuration API to provision the device's initial configuration — including firmware version, port settings, and operational policies captured as `activationFeature` items. The Monitor resource is used to track the asynchronous completion of the provisioning workflow.

Update resource configuration after maintenance (Use Case 3)

Following a planned maintenance window, the operations system uses the PATCH operation on the Resource Activation and Configuration API to update the `administrativeState` back to `unlocked`, `operationalState` to `enabled`, and `usageState` to `active`. A `ResourceStateChangeEvent` notification is published to subscribing systems to signal the return to service.

Decommission a resource at end of life (Use Case 4)

When a physical device reaches its end of life or a logical number is no longer required, the Resource Activation and Configuration API is called to set `administrativeState` to `locked` and `operationalState` to `disabled` before the resource record is permanently deleted. A `ResourceDeleteEvent` notification is published to allow downstream inventory and billing systems to reconcile their records.

Asynchronous resource activation monitoring (Use Case 5)

For resource activation requests that involve long-running or external processes (e.g., provisioning a circuit across multiple domains), the API creates a Monitor resource linked to

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the original request. The client polls the Monitor or receives a `MonitorStateChangeEvent` notification when the activation completes or fails. The Monitor records both the original request and the system's response for audit purposes.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a PhysicalResource and LogicalResource inheriting properties from the abstract Resource entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Resource instances some may be instances of PhysicalResource where other could be instances of LogicalResource. The @type gives this information. All resources and sub-resources of this API have a @type attributes that must be provided.

The @referredType can be used within reference entities (like for instance an ResourceRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type denotes the class type of the reference itself, such as PhysicalResourceRef or LogicalResourceRef, and not the class type of the referred object.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

Resource resource

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

Resource model

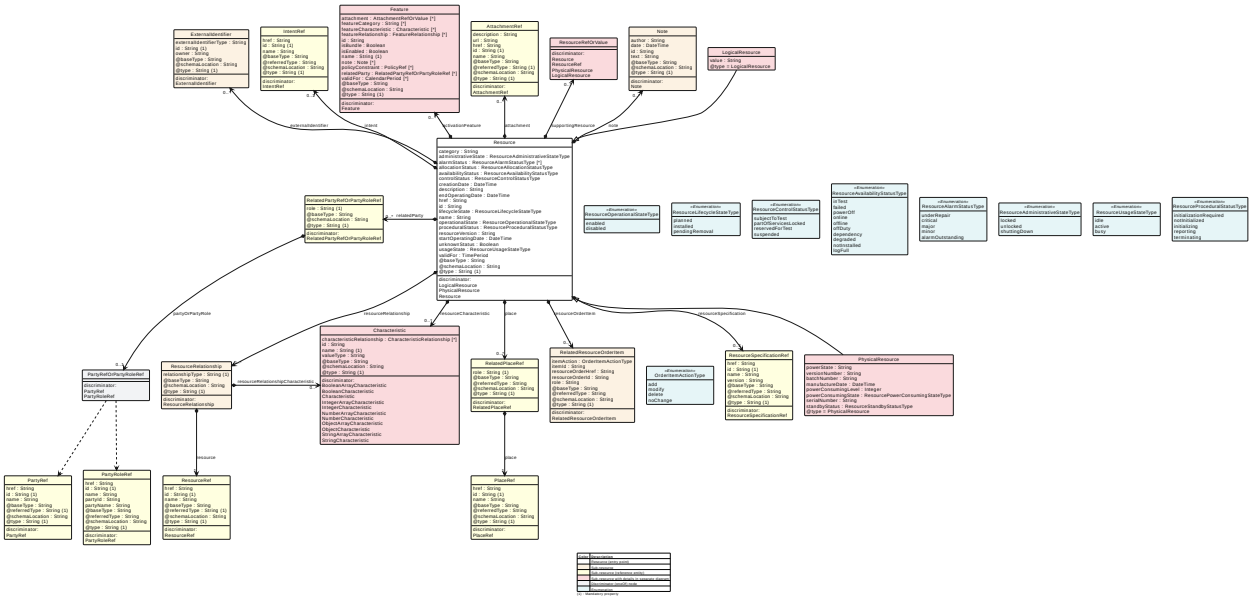


Figure 1 - Resource

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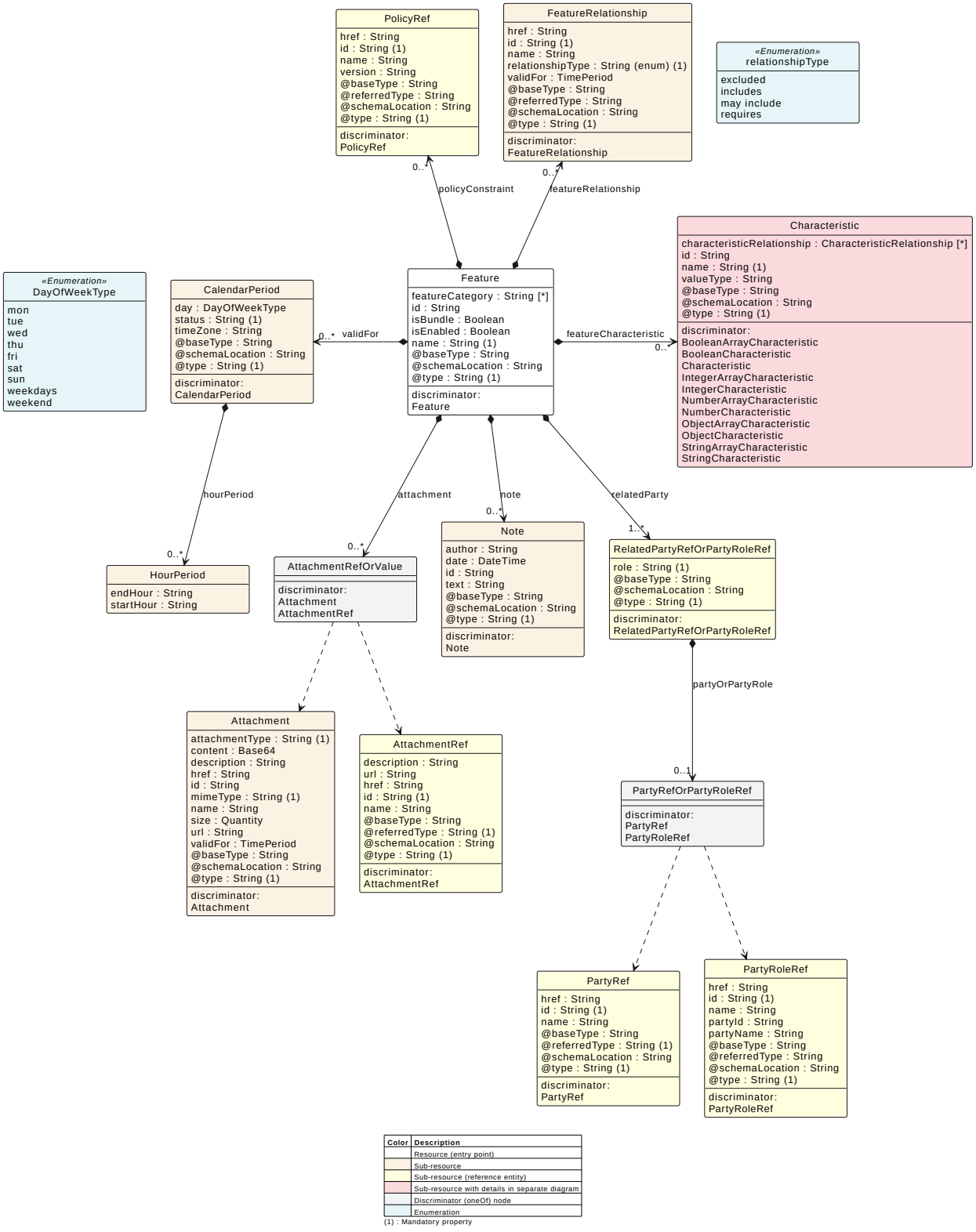


Figure 2 - Feature

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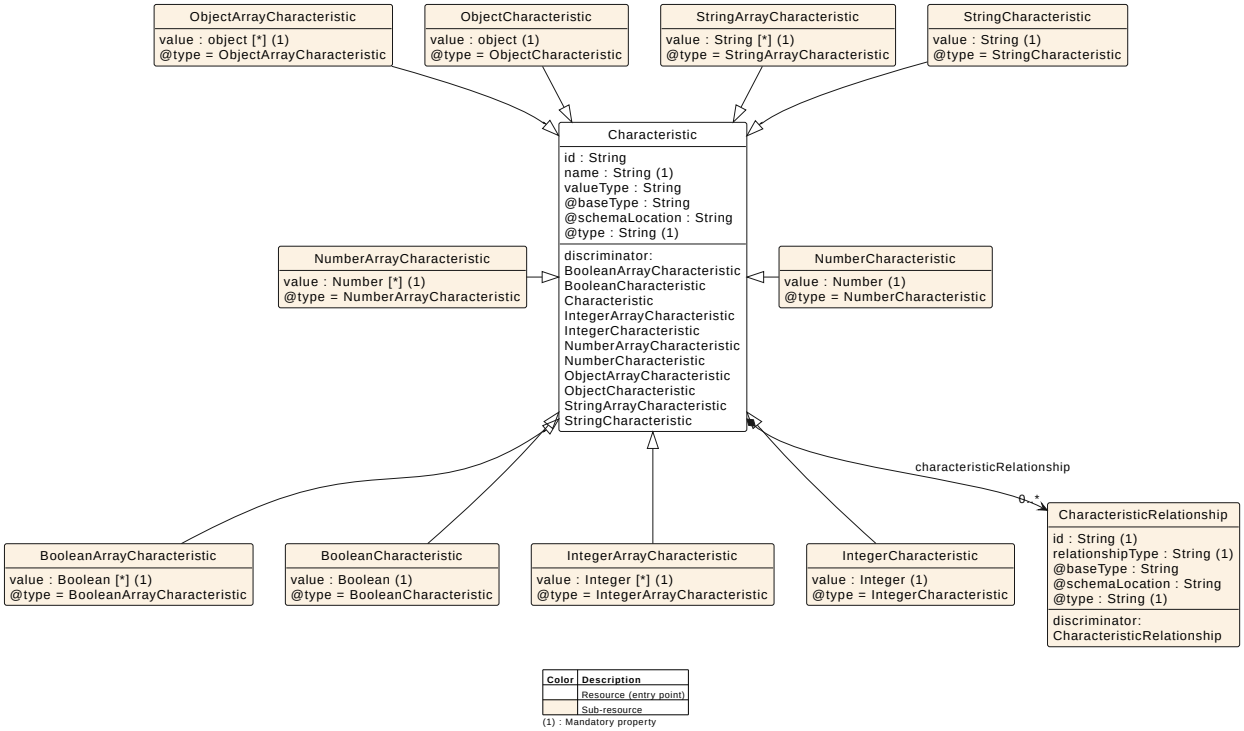


Figure 3 - Characteristic

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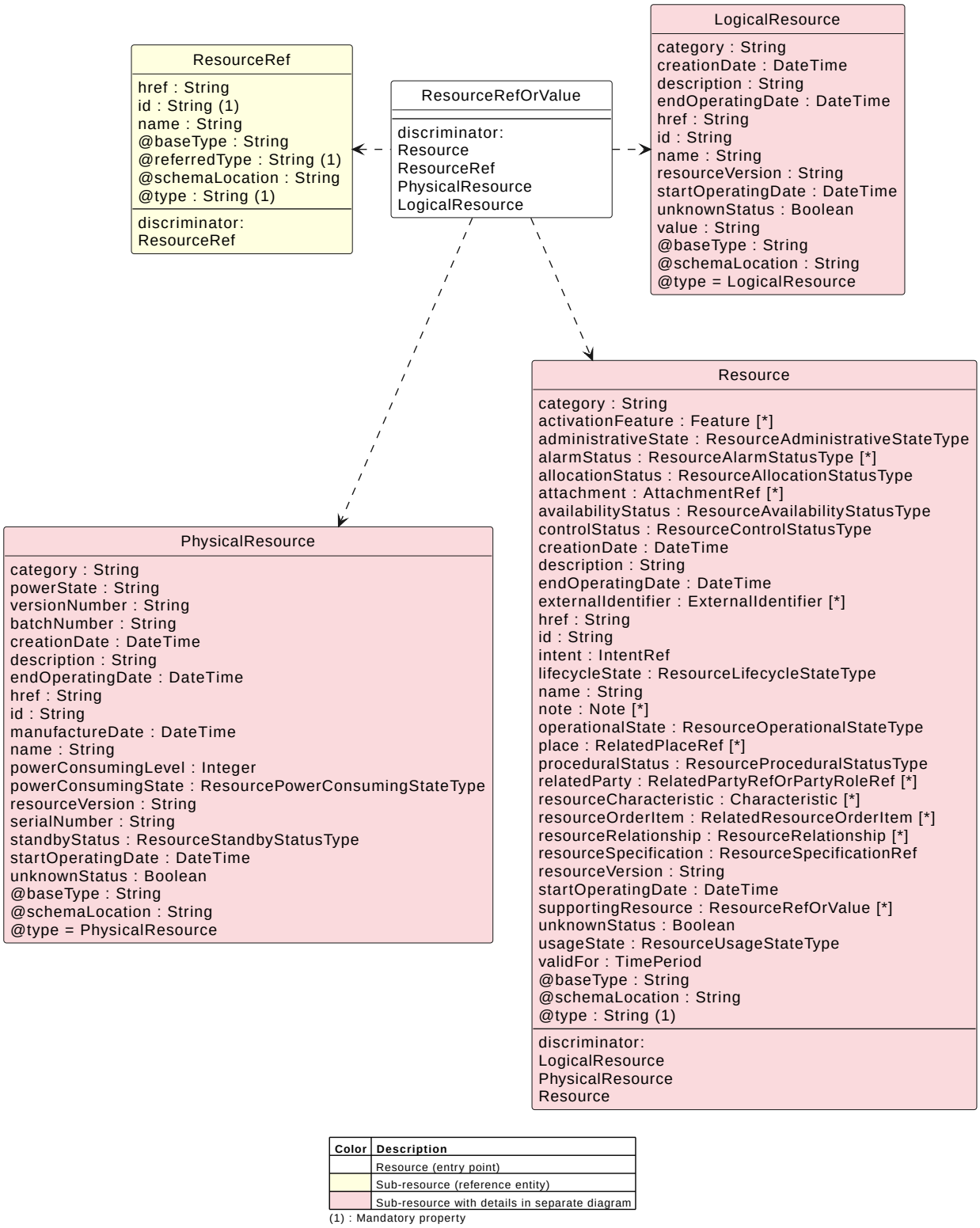


Figure 4 - ResourceRefOrValue

Field descriptions

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Resource fields

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

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externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

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place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

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relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

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Configuration feature.

attachment	An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.
featureCategory	A String [0..*]. Collection of feature categories.
featureCharacteristic	A Characteristic [0..*]. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship [0..*]. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String [1]. The name for the feature.
note	A Note [0..*]. Notes relevant to this Feature.
policyConstraint	A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.
validFor	A CalendarPeriod [0..*]. Validity period of this Feature.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

attachmentType	A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, embedded inline and encoded as base64. Mutually exclusive with url.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String [1]. A technical characterization of the attachment content format using an IETF Media Type, for example application/pdf or image/png. Note: this property is named mimeType for historical reasons.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. The URL pointing to the attachment content held in an external system, such as a content management system or document repository. Mutually exclusive with content.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

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~~description~~	DEPRECATED A String. A narrative text describing the content of the attachment. DEPRECATED: will be removed in v6.
href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
~~url~~	DEPRECATED A String. Link to the attachment media/content. DEPRECATED: will be removed in v6.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

id	A String [1]. Unique identifier of the characteristic.
relationshipType	A String [1]. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1..*]. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Boolean [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1..*]. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	An Integer [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1..*]. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

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A characteristic which value is a Number.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A Number [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s). If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType [1..*]. A characteristic which value is an array of Object(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

A characteristic which value is an Object. If this characteristic is not specified by a

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CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A valueType_2 [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1..*]. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

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characteristicRelationship	A CharacteristicRelationship [0..*]. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String [1]. Name of the characteristic.
value	A String [1]. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

Configuration feature.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. This is the name of the target feature.
relationshipType	A String [1]. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

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Extra information about a given entity.

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

Reference to managed Policy object.

href	A String. Hyperlink reference.
id	A String [1]. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.

partyOrPartyRole	A PartyRefOrPartyRoleRef. A reference to a Party or a PartyRole.

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role	A String [1]. The role the referred Party or PartyRole plays in the specific context of the entity it is linked to, such as 'initiator', 'customer', 'salesAgent', 'user'; NOTE: If used to refer to a PartyRole, the PartyRole describes the role the respective engagedParty plays in a more general context, like e.g. 'Customer', 'Manager'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

A Party reference.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String [1]. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRef

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.

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partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRoleRef

CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

day	A DayOfWeekType. Possible values for the days of the week. ENUMERATED with values: * mon * tue * wed * thu * fri * sat * sun * weekdays * weekend
hourPeriod	A HourPeriod [0..*]. Start and end time range in 24-hour HH:mm format.
status	A String [1]. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information in ISO 8601 timezone: either 'Z' for UTC or an offset like +02:00 or -05:30.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

Hour interval.

endHour	A String. The end of the hour period, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight).
startHour	A String. The beginning of the hour period, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM).

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String [1]. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

IntentRef sub-resource fields

Intent reference, for when Intent is used by other entities.

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href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRef sub-resource fields

Use role to distinguish the context of the place being referenced (such as `delivery-entrance`). Use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). Use `@type` to extend (eg: as RefOrValue).

place	A PlaceRef [1]. A reference to a physical Place.
role	A String [1]. The context in which this Place is being used.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The kind of Place that is being referenced.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

A reference to a physical Place.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The specific sub-type of Place being referred to (such as a `GeographicAddress`).

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedResourceOrderItem sub-resource fields

RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.

itemAction	An OrderItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values: * add * modify * delete * noChange
itemId	A String. Identifier of the order item where the resource was managed.
resourceOrderHref	A String. Reference of the related entity.
resourceOrderId	A String. Unique identifier of a related entity.
role	A String. Role of the resource order item for this resource.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceRelationship sub-resource fields

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

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relationshipType	A String [1]. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [relies On] if the resource needs another already owned resource to rely on (eg: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A ResourceRef [1]. Resource reference, for when Resource is used by other entities.
resourceRelationshipCharacteristic	A Characteristic [0..*]. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic... Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

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href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PhysicalResource sub-resource fields

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Physical resource is a type of resource that describes the common set of attributes shared by all concrete physical resources (e.g. EQUIPMENT) in the inventory.

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

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batchNumber	A String. A unique identifier assigned to a specific group of physical resources produced or received under the same conditions for tracking and traceability in the supply chain.
~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.

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endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
manufactureDate	A DateTime. This is a string attribute that defines the date of manufacture of this item in the ISO 8601 date time standard format. This is an optional attribute.
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.

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operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
powerConsumingLevel	An Integer. The power consuming level attribute (defined by ITU-T M.3701) indicates the current percentage of power consuming of the resource represented by the managed object, where 100 percent represents the maximum power consumption of the device. It is single valued and read-only by management functions.

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powerConsumingState	A ResourcePowerConsumingStateType. Value enumeration for the 'resourcePowerConsumingState' of a Resource; the power consuming state attribute (defined by ITU-T M.3701) indicates the power consuming policy of the resource represented by the managed object, it is single valued and read-write by management functions.; defined values: 'fullPower': The resource runs in full power and will not consider power saving. If a large device contains parts that also support this attribute, when the container object set to full power state, all its parts shall turn into full power state; 'powerSaving': The resource is (or may be) partially running to save power; depending on how much the payload is lower than the maximum capacity of the resource, the degree of saving will be automatically decided by the resource. If power saving is not supported by the resource, attempting to change to this state will have the same effect as full power state. If a large device contains parts that also support this attribute, when the container object is set to power saving state, its parts may still remain in full power state. But if one of the parts turns into power saving state, the container object shall turn into power saving state; 'sleeping': The resource is not running but keeps a snapshot of its memory before sleeping, and can resume running mode quickly. In this mode, the power is only consumed to meet the need of maintaining the memory. If it is not supported by the resource, attempting to change to this state will result in a failure; 'shutdown': The resource has been powered off and does not consume any more power; if it is not supported by the resource, attempting to change to this state will result in a failure. ENUMERATED with values: * fullPower * powerSaving * sleeping * shutdown

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~~powerState~~	DEPRECATED A String. This defines the current power status of the hardware item. Values include: 0: Unknown 1: Not Applicable 2: No Power Applied 3: Full Power Applied 4: Power Save - Normal 5: Power Save - Degraded 6: Power Save - Standby 7: Power Save - Critical 8: Power Save - Low Power Mode 9: Power Save - Unknown 10: Power Cycle 11: Power Warning 12: Power Off; NOTE: This attribute is replaced by the 'powerConsuming State' attribute, to align with the official name as defined by ITU-T.

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).

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resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.

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serialNumber	A String. This is a string that represents a manufacturer-allocated number used to identify different instances of the same hardware item. The ModelNumber and Part Number attributes are used to identify different types of hardware items.
standbyStatus	A ResourceStandbyStatusType. Value enumeration for the 'resourceStandbyStatus' of a Resource; the standby status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the working mode for standby objects, i.e. whether it is providing service or standby as a back-up resource; it is single valued and read-only by management functions; defined values: 'hotStandby': The resource is not providing service, but is operating in synchronism with another resource that is to be backed-up (e.g., a computer shadowing another computer). A resource with a hot standby status will be immediately able to take over the role of the resource to be backed up, without the need for initialization activity, and will contain the same information as the resource to be backed up; 'coldStandby': The resource is to back up another resource, but is not synchronized with that resource. A resource with a cold standby status will not be immediately able to take over the role of a resource to be backed up, and will require some initialization activity; 'providingService': The back up resource is providing service and is backing up another resource. ENUMERATED with values: * hotStandby * coldStandby * providingService
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.

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usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
~~versionNumber~~	DEPRECATED A String. This is a string that identifies the version of this physical resource. This is an optional attribute. NOTE that this attribute is deprecated as it overlaps with the Resource.resourceVersion attribute.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

LogicalResource sub-resource fields

Logic resource is a type of resource that describes the common set of attributes shared by all concrete logical resources (e.g. TPE, MSISDN, IP Addresses) in the inventory.

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.

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administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

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externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

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place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

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relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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Json representation sample(s)

Example LogicalResource (MSISDN) shown under the abstract Resource type for documentation tooling. Concrete PhysicalResource and LogicalResource examples are declared separately on each subclass.


```
{
  "@type": "LogicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
  "id": "8044",
  "category": "Mobile",
  "description": "A mobile subscriber number (MSISDN) assigned to a customer.
Represents the logical resource used to identify a subscriber in the GSM/LTE network.",
  "name": "MSISDN +44-7700-900123",
  "startOperatingDate": "2023-01-15T00:00:00Z",
  "endOperatingDate": "2026-01-15T00:00:00Z",
  "administrativeState": "unlocked",
  "operationalState": "enabled",
  "usageState": "active",
  "validFor": {
    "startDateTime": "2023-01-15T00:00:00Z",
    "endDateTime": "2026-01-15T00:00:00Z"
  },
  "resourceVersion": "1.0.0",
  "note": [
    {
      "@type": "Note",
      "id": "note-001",
      "author": "network-provisioning-system",
      "date": "2023-01-15T08:30:00Z",
      "text": "MSISDN provisioned and activated on LTE network. Ported from previous
operator."
    }
  ],
  "resourceOrderItem": [
    {
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem",
      "resourceOrderHref": "/tmf-api/resourceOrderManagement/v5/resourceOrder/ORD-2023-
0042",
      "resourceOrderId": "ORD-2023-0042",
      "itemAction": "add",
      "itemId": "1"
    }
  ],
  "place": [
    {
      "@type": "RelatedPlaceRef",
      "role": "RegisteredAddress",
      "place": {
        "id": "addr-9912",
        "href": "/tmf-api/geographicAddressManagement/v5/geographicAddress/addr-9912",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress",
        "name": "10 Downing Street, London, SW1A 2AA, UK"
      }
```


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```
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "user",
        "partyOrPartyRole": {
          "@type": "PartyRoleRef",
          "href": "/tmf-api/partyManagement/v5/partyRole/456",
          "id": "456",
          "name": "Jane Smith",
          "@referredType": "Employee"
        }
      }
    ],
    "supportingResource": [
      {
        "@type": "ResourceRef",
        "id": "545",
        "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/545",
        "name": "SIM Card ICCID-8944110012345678901",
        "@referredType": "PhysicalResource"
      }
    ],
    "resourceRelationship": [
      {
        "@type": "ResourceRelationship",
        "relationshipType": "reliesOn",
        "resource": {
          "@type": "ResourceRef",
          "id": "34",
          "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/34",
          "name": "LTE Base Station Node-B-23",
          "@referredType": "PhysicalResource"
        },
        "resourceRelationshipCharacteristic": [
          {
            "@type": "NumberCharacteristic",
            "@baseType": "Characteristic",
            "id": "char-223",
            "name": "signalStrength",
            "valueType": "number",
            "value": -75
          }
        ]
      }
    ],
    "resourceCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "@baseType": "Characteristic",
        "id": "char-001",
        "name": "msisdn",
        "valueType": "string",
        "value": "+447700900123"
      },
      {
        "@type": "StringCharacteristic",
        "@baseType": "Characteristic",
        "id": "char-002",
```


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```
      "name": "networkType",
      "valueType": "string",
      "value": "LTE"
    },
    {
      "@type": "NumberCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-003",
      "name": "portingDays",
      "valueType": "number",
      "value": 3
    }
  ],
  "attachment": [
    {
      "@type": "AttachmentRef",
      "href": "/tmf-api/documentManagement/v5/attachment/att-22",
      "id": "att-22",
      "name": "Service Activation Certificate",
      "@referredType": "Attachment"
    }
  ],
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-101",
    "id": "spec-101",
    "name": "MSISDN",
    "@referredType": "ResourceSpecification",
    "version": "2.1.0"
  },
  "activationFeature": [
    {
      "@type": "Feature",
      "id": "feat-001",
      "name": "VoLTE",
      "isBundle": false,
      "isEnabled": true,
      "featureCharacteristic": [
        {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "id": "char-feat-001",
          "name": "codecProfile",
          "valueType": "string",
          "value": "AMR-WB"
        }
      ],
      "featureRelationship": [
        {
          "@type": "FeatureRelationship",
          "id": "feat-rel-001",
          "name": "requires",
          "relationshipType": "requires",
          "validFor": {
            "startDateTime": "2023-01-15T00:00:00Z",
            "endDateTime": "2026-01-15T00:00:00Z"
          }
        }
      ],
      "policyConstraint": [
```


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```
        {
          "@type": "PolicyRef",
          "href": "/tmf-api/policyManagement/v5/policy/pol-012",
          "id": "pol-012",
          "name": "VoLTE QoS Policy",
          "@referredType": "Policy",
          "version": "1.0.0"
        }
      ]
    }
  ],
  "intent": {
    "@type": "IntentRef",
    "href": "/tmf-api/intentManagement/v5/intent/int-124",
    "id": "int-124",
    "name": "Ensure 99.9% network availability",
    "@referredType": "Intent"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "ext-001",
      "externalIdentifierType": "IMSI",
      "owner": "HomeNetworkOperator"
    }
  ]
}
```

Monitor resource

A mechanism used to track activation and configuration of Services and Resources. Specifically for Resource Functions, a Monitor can be used to track the progress of activation or of other tasks related to resource functions (such as healing, scaling, migration).

Resource model

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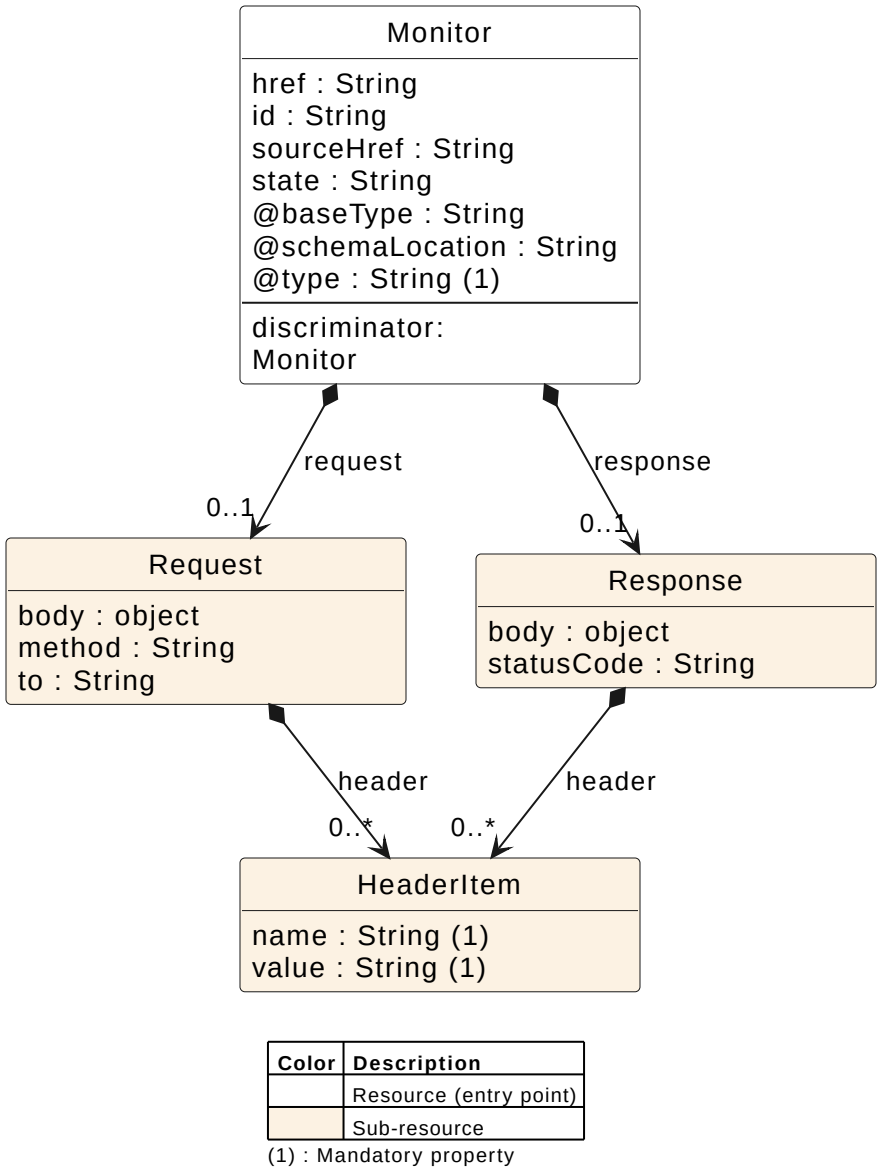


Figure 5 - Monitor

Field descriptions

Monitor fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
request	A Request. A response to a request.
response	A Response. A response to a request.
sourceHref	A String. The monitored resource href.

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state	A String. The Monitor state of the resource. inProgress, in Error, completed.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Request sub-resource fields

A response to a request.

body	A bodyType. The body of the request. For example for an HTTP request might contain content of a form .
header	A HeaderItem [1..*]. Items included in the header of the request. For example for an HTTP request might contain requested locale, basic authentication.
method	A String. The protocol of the request, e.g. http.
to	A String. The target of the request, e.g. a URL for an HTTP request.

HeaderItem sub-resource fields

An item typically included in a request or response.

name	A String [1]. The name of the header item, e.g. locale.
value	A String [1]. The value of the header item, e.g. en-us.

Response sub-resource fields

A response to a request.

body	A bodyType_2. The body of the response.
header	A HeaderItem [1..*]. Items included in the header of the response. For example for an HTTP response might contain negotiated locale.
statusCode	A String. The status of the response. For example for an HTTP response would be codes such as 200, 400, etc.

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Json representation sample(s)

Example Monitor resource correlating an HTTP request/response pair for a PhysicalResource activation under Resource Activation and Configuration.


```
{
  "@type": "Monitor",
  "@schemaLocation": "/schemaRegistry/Monitor.yaml",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/Monitor/mon-1007",
  "id": "mon-1007",
  "request": {
    "method": "POST",
    "to": "/tmf-api/resourceActivationAndConfiguration/v5/resource/res-8044",
    "body": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "@schemaLocation": "/schemaRegistry/LogicalResource.schema.json",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/res-8044",
      "id": "res-8044",
      "category": "Mobile",
      "description": "MSISDN logical resource submitted for activation on the
network.",
      "name": "MSISDN +1-555-0100999",
      "endOperatingDate": "2026-01-15T00:00:00Z",
      "administrativeState": "unlocked",
      "operationalState": "enabled",
      "usageState": "active",
      "validFor": {
        "startDateTime": "2024-11-19T06:36:48Z",
        "endDateTime": "2026-11-19T06:36:48Z"
      }
    },
    "header": [
      {
        "name": "Content-Type",
        "value": "application/json"
      },
      {
        "name": "Accept",
        "value": "application/json"
      }
    ]
  },
  "response": {
    "statusCode": "200",
    "body": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "@schemaLocation": "/schemaRegistry/LogicalResource.schema.json",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/res-8044",
      "id": "res-8044",
      "category": "Mobile",
      "description": "MSISDN logical resource after activation monitoring shows an
alarm state on a dependent check.",
      "name": "MSISDN +1-555-0100999",
      "endOperatingDate": "2026-01-15T00:00:00Z",
      "administrativeState": "unlocked",
```


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```
      "operationalState": "enabled",
      "usageState": "idle",
      "validFor": {
        "startDateTime": "2024-11-19T06:36:48Z",
        "endDateTime": "2026-11-19T06:36:48Z"
      }
    },
    "header": [
      {
        "name": "Content-Type",
        "value": "application/json"
      },
      {
        "name": "Cache-Control",
        "value": "no-cache"
      }
    ]
  },
  "sourceHref": "/tmf-api/resourceActivationAndConfiguration/v5/resource/res-8044",
  "state": "completed"
}
```


PhysicalResource resource

Physical resource is a type of resource that describes the common set of attributes shared by all concrete physical resources (e.g. EQUIPMENT) in the inventory.

Resource model

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Figure 6 - PhysicalResource

Field descriptions

PhysicalResource fields

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.

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administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to

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batchNumber	A String. A unique identifier assigned to a specific group of physical resources produced or received under the same conditions for tracking and traceability in the supply chain.
~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.

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endOperatingDate	A DateTime. The point in time till the resource is operating.
externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
manufactureDate	A DateTime. This is a string attribute that defines the date of manufacture of this item in the ISO 8601 date time standard format. This is an optional attribute.
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.

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operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled
place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
powerConsumingLevel	An Integer. The power consuming level attribute (defined by ITU-T M.3701) indicates the current percentage of power consuming of the resource represented by the managed object, where 100 percent represents the maximum power consumption of the device. It is single valued and read-only by management functions.

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powerConsumingState	A ResourcePowerConsumingStateType. Value enumeration for the 'resourcePowerConsumingState' of a Resource; the power consuming state attribute (defined by ITU-T M.3701) indicates the power consuming policy of the resource represented by the managed object, it is single valued and read-write by management functions.; defined values: 'fullPower': The resource runs in full power and will not consider power saving. If a large device contains parts that also support this attribute, when the container object set to full power state, all its parts shall turn into full power state; 'powerSaving': The resource is (or may be) partially running to save power; depending on how much the payload is lower than the maximum capacity of the resource, the degree of saving will be automatically decided by the resource. If power saving is not supported by the resource, attempting to change to this state will have the same effect as full power state. If a large device contains parts that also support this attribute, when the container object is set to power saving state, its parts may still remain in full power state. But if one of the parts turns into power saving state, the container object shall turn into power saving state; 'sleeping': The resource is not running but keeps a snapshot of its memory before sleeping, and can resume running mode quickly. In this mode, the power is only consumed to meet the need of maintaining the memory. If it is not supported by the resource, attempting to change to this state will result in a failure; 'shutdown': The resource has been powered off and does not consume any more power; if it is not supported by the resource, attempting to change to this state will result in a failure. ENUMERATED with values: * fullPower * powerSaving * sleeping * shutdown

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~~powerState~~	DEPRECATED A String. This defines the current power status of the hardware item. Values include: 0: Unknown 1: Not Applicable 2: No Power Applied 3: Full Power Applied 4: Power Save - Normal 5: Power Save - Degraded 6: Power Save - Standby 7: Power Save - Critical 8: Power Save - Low Power Mode 9: Power Save - Unknown 10: Power Cycle 11: Power Warning 12: Power Off; NOTE: This attribute is replaced by the 'powerConsumingState' attribute, to align with the official name as defined by ITU-T. +

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proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating
relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).

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resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.

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serialNumber	A String. This is a string that represents a manufacturer-allocated number used to identify different instances of the same hardware item. The ModelNumber and Part Number attributes are used to identify different types of hardware items.
standbyStatus	A ResourceStandbyStatusType. Value enumeration for the 'resourceStandbyStatus' of a Resource; the standby status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the working mode for standby objects, i.e. whether it is providing service or standby as a back-up resource; it is single valued and read-only by management functions; defined values: 'hotStandby': The resource is not providing service, but is operating in synchronism with another resource that is to be backed-up (e.g., a computer shadowing another computer). A resource with a hot standby status will be immediately able to take over the role of the resource to be backed up, without the need for initialization activity, and will contain the same information as the resource to be backed up; 'coldStandby': The resource is to back up another resource, but is not synchronized with that resource. A resource with a cold standby status will not be immediately able to take over the role of a resource to be backed up, and will require some initialization activity; 'providingService': The back up resource is providing service and is backing up another resource. ENUMERATED with values: * hotStandby * coldStandby * providingService
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.

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usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
~~versionNumber~~	DEPRECATED A String. This is a string that identifies the version of this physical resource. This is an optional attribute. NOTE that this attribute is deprecated as it overlaps with the Resource.resourceVersion attribute.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

Configuration feature.

See [Feature] for details.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

See [Attachment] for details.

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AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

See [AttachmentRef] for details.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

See [IntegerCharacteristic] for details.

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NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s). If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

See [ObjectArrayCharacteristic] for details.

valueType sub-resource fields

Characteristic value.

See [valueType] for details.

ObjectCharacteristic sub-resource fields

A characteristic which value is an Object. If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

See [ObjectCharacteristic] for details.

valueType_2 sub-resource fields

Value of the characteristic.

See [valueType_2] for details.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

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See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

See [StringCharacteristic] for details.

FeatureRelationship sub-resource fields

Configuration feature.

See [FeatureRelationship] for details.

Note sub-resource fields

Extra information about a given entity.

See [Note] for details.

PolicyRef sub-resource fields

Reference to managed Policy object.

See [PolicyRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

A Party reference.

See [PartyRef] for details.

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

See [PartyRoleRef] for details.

CalendarPeriod sub-resource fields

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The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

See [CalendarPeriod] for details.

HourPeriod sub-resource fields

Hour interval.

See [HourPeriod] for details.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

See [ExternalIdentifier] for details.

IntentRef sub-resource fields

Intent reference, for when Intent is used by other entities.

See [IntentRef] for details.

RelatedPlaceRef sub-resource fields

Use role to distinguish the context of the place being referenced (such as `delivery-entrance`). Use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). Use `@type` to extend (eg: as RefOrValue).

See [RelatedPlaceRef] for details.

PlaceRef sub-resource fields

A reference to a physical Place.

See [PlaceRef] for details.

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RelatedResourceOrderItem sub-resource fields

RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.

See [RelatedResourceOrderItem] for details.

ResourceRelationship sub-resource fields

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

See [ResourceRelationship] for details.

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

See [ResourceRef] for details.

ResourceSpecificationRef sub-resource fields

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

See [ResourceSpecificationRef] for details.

Resource sub-resource fields

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

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activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to no service at all. Object specific attributes may be defined

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

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externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

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place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

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relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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	Resource can be instantiated as * LogicalResource * PhysicalResource
value	This property is present in subclasses

LogicalResource sub-resource fields

Logic resource is a type of resource that describes the common set of attributes shared by all concrete logical resources (e.g. TPE, MSISDN, IP Addresses) in the inventory.

See [LogicalResource] for details.

Json representation sample(s)

Example PhysicalResource representing routed network equipment used in activation tests.


```
{
  "@type": "PhysicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-2201",
  "id": "pr-2201",
  "category": "NetworkEquipment",
  "description": "Cisco ASR 9000 series router installed at the London Docklands data
centre. Provides core IP routing for the metropolitan area network.",
  "name": "ASR9001-LDNDC-01",
  "startOperatingDate": "2022-06-01T00:00:00Z",
  "endOperatingDate": "2027-06-01T00:00:00Z",
  "administrativeState": "unlocked",
  "operationalState": "enabled",
  "usageState": "active",
  "resourceVersion": "1.0.0",
  "serialNumber": "FCZ2150R1GH",
  "manufactureDate": "2021-11-10T00:00:00Z",
  "powerState": "on",
  "versionNumber": "IOS-XR 7.5.1",
  "resourceCharacteristic": [
    {
      "@type": "NumberCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-pr-001",
      "name": "portCount",
      "valueType": "number",
      "value": 48
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-pr-002",
      "name": "rackUnit",
      "valueType": "string",
      "value": "U14"
```


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```
      }
    ],
    "resourceSpecification": {
      "@type": "ResourceSpecificationRef",
      "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-202",
      "id": "spec-202",
      "name": "Cisco ASR 9001",
      "@referredType": "ResourceSpecification",
      "version": "1.0.0"
    }
  }
```

LogicalResource resource

Logic resource is a type of resource that describes the common set of attributes shared by all concrete logical resources (e.g. TPE, MSISDN, IP Addresses) in the inventory.

Resource model

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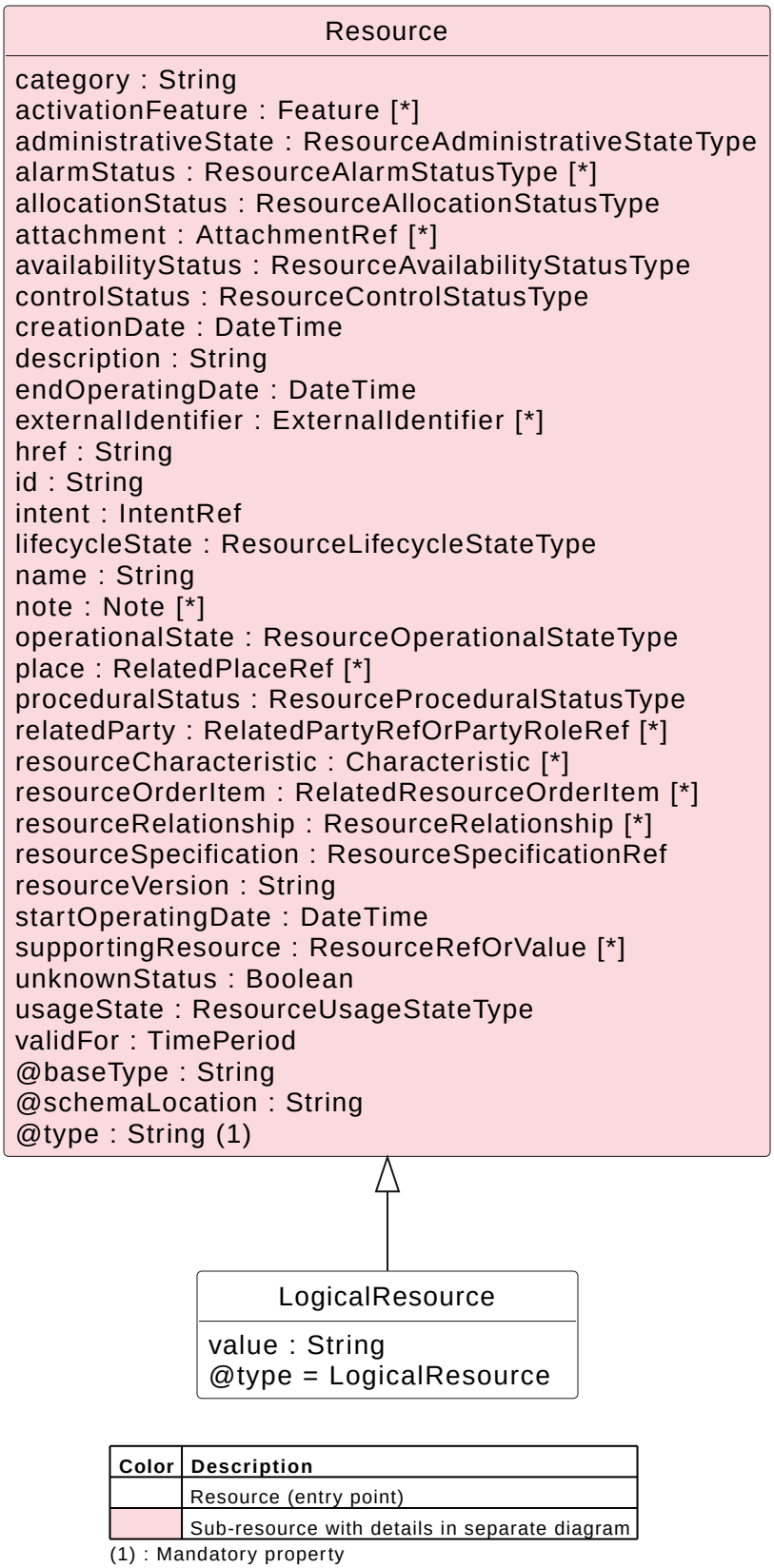


Figure 7 - LogicalResource

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Field descriptions

LogicalResource fields

activationFeature	A Feature [0..*]. Configuration features; see TMF664 Resource Function Activation API.
administrativeState	A ResourceAdministrativeStateType. Value enumeration for the 'resourceAdministrativeState' of a Resource; the administrative state attribute (defined by ITU-T X.731/ITU-T M.3701) represents permission to use ('unlocked') or prohibition against using ('locked') the resource imposed through the management services; it is single valued and read-write by management functions; defined values: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users; this is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shuttingDown

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alarmStatus	A ResourceAlarmStatusType [0..*]. Value enumeration for the 'resourceAlarmStatus' of a Resource; the alarm status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the severity status of fault for a resource; it is multi-valued and read-only by management functions, each value represents an existing severity; defined values: 'underRepair': The resource is currently being repaired; 'critical': One or more critical alarms indicating a fault have been detected in the resource, and have not been cleared; 'major': One or more major alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'minor': One or more minor alarms indicating a fault have been detected in the resource, and have not yet been cleared; 'alarmOutstanding': One or more alarms have been detected in the resource. The condition may or may not be disabling. If the operational state is 'enabled', additional attributes, particular to the managed object class, may indicate the nature and cause of the condition and the services that are affected. ENUMERATED with values: * underRepair * critical * major * minor * alarmOutstanding
allocationStatus	A ResourceAllocationStatusType. Values for the 'allocationStatus' of a Resource; for example: 'available', 'reserved', 'allocated' and 'partiallyAllocated'.
attachment	An AttachmentRef [0..*]. A list of references to attachments associated with this resource.

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availabilityStatus	A ResourceAvailabilityStatusType. Value enumeration for the 'resourceAvailabilityStatus' of a Resource; the availability status attribute (defined by ITU-T X.731/ITU-T M.3701) represents the availability of the resource for either carrying service or for being in other states, such as in testing; it is single valued and read-only by management functions; defined values: 'inTest': The resource is undergoing a test procedure. If the administrative state is 'locked' or 'shuttingDown', then normal users are precluded from using the resource, and the controlStatus attribute has the value 'reservedForTest'. Tests that do not exclude additional users can be present in any operational or administrative state but the 'reservedForTest' condition should not be present; 'failed': The resource has an internal fault that prevents it from operating. The operational state is 'disabled'; 'powerOff': The resource requires power to be applied and is not powered on. The operational state is 'disabled'; 'online': The resource is available for use. The operational state is 'enabled'; 'offline': The resource requires a routine operation to be performed to place it on line and make it available for use. The operation may be manual or automatic, or both. The operational state is 'disabled'; 'offDuty': The resource has been made inactive by an internal control process in accordance with a predetermined time schedule. Under normal conditions the control process can be expected to reactivate the resource at some scheduled time, and it is therefore considered to be optional. The operational state is 'enabled' or 'disabled'; 'dependency': The resource cannot operate because some other resource on which it depends (i.e., a resource not represented by the same managed object) is unavailable. The operational state is 'disabled'; 'degraded': The service available from the resource is degraded in some respect, such as in speed or operating capacity. Failure of a test or an unacceptable performance measurement has established that some or all services are not functional or are degraded due to the presence of a defect. However, the resource remains available for service, either because some services are satisfactory or because degraded service is preferable to

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~~category~~	DEPRECATED A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource; NOTE: this property is misleading and therefore is deprecated.
controlStatus	A ResourceControlStatusType. Value enumeration for the 'resourceControlStatus' of a Resource; the control status attribute (defined by ITU-T X.731/ITU-T M.3701) indicates that the resource is under some kind of control; it is single valued and read-only by management functions; defined values: 'subjectToTest': The resource is available to normal users, but tests may be conducted on it simultaneously at unpredictable times, which may cause it to exhibit unusual characteristics to users; 'partOfServicesLocked': This value indicates whether a manager has administratively restricted a particular part of a service from the user(s) of a resource. The administrative state is 'unlocked'. Examples are incoming service barred, outgoing service barred, write locked by media key, read locked; 'reservedForTest': The resource has been made administratively unavailable to normal users because it is undergoing a test procedure. The administrative state is 'locked'; 'suspended': Service has been administratively suspended to the users of the resource. The resource may retain knowledge of current users and/or requests for usage, depending upon the managed object class definition, but does not resume performing services until the suspended condition is revoked. The administrative state is 'unlocked'. ENUMERATED with values: * subjectToTest * partOfServicesLocked * reservedForTest * suspended
creationDate	A DateTime. The point in time when the resource was created.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. The point in time till the resource is operating.

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externalIdentifier	An ExternalIdentifier [0..*]. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
lifecycleState	A ResourceLifecycleStateType. Value enumeration for the 'resourceLifecycleState' of a Resource; the lifecycle state attribute (defined by ITU-T X.731/ITU-T M.3701) represents the planned and tracked state of a physical or logical resource; it is single valued and read-write by management functions; defined values: 'planned', 'installed', 'pendingRemoval'. ENUMERATED with values: * planned * installed * pendingRemoval
name	A String. The name of the resource.
note	A Note [0..*]. A list of notes associated with this resource.
operationalState	A ResourceOperationalStateType. Value enumeration for the 'resourceOperationalState' of a Resource; the operational state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether or not the resource is installed and working; it is single valued and read-only by management functions; defined values: 'disabled': The resource is totally inoperable and unable to provide service to the user(s); 'enabled': The resource is partially or fully operable and available for use. ENUMERATED with values: * enabled * disabled

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place	A RelatedPlaceRef [0..*]. A list of references to places related to this Resource, differentiated by their role (e.g., installation place, usage place).
proceduralStatus	A ResourceProceduralStatusType. Value enumeration for the 'resourceProceduralStatus' of a Resource; the procedural status attribute (defined by ITU-T X.731/ITU-T M.3701) represents a sequence of phases through which the resource needs to progress; it is single valued and read-only by management functions; defined values: 'initializationRequired': The resource requires initialization to be invoked by the manager before it can perform its normal functions, and this procedure has not been initiated. The manager may be able to invoke such initialization through an action. The operational state is 'disabled'; 'notInitialized': The resource requires initialization before it can perform its normal functions, and this procedure has not been initiated. The resource initializes itself autonomously, but the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition; 'initializing': The resource requires initialization before it can perform its normal functions, and this procedure has been initiated but is not yet complete. The operational state may be 'disabled' or 'enabled', depending upon the managed object class definition; 'reporting': The resource has completed some processing operation and is notifying the results of the operation, e.g., a test process is sending its results. The operational state is 'enabled'; 'terminating': The resource is in a termination phase. If the resource does not reinitialize itself autonomously, the operational state is 'disabled'. Otherwise, the operational state may be either 'disabled' or 'enabled', depending upon the managed object class definition. ENUMERATED with values: * initializationRequired * notInitialized * initializing * reporting * terminating

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relatedParty	A RelatedPartyRefOrPartyRoleRef [0..*]. A list of Parties or PartyRoles related to this resource, differentiated by their role (e.g., owner, user, installer).
resourceCharacteristic	A Characteristic [0..*]. A list of Characteristics of this resource. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
resourceOrderItem	A RelatedResourceOrderItem [0..*]. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship [0..*]. A list of relationships between this resource and other resources, differentiated by relationshipType.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceVersion	A String. A field that identifies the specific version of an instance of a resource. NOTE: At some point in the future (v6) this will be aligned with the new Version pattern, and then this property would need to be renamed to **version**.
startOperatingDate	A DateTime. The point in time from which the resource is operating.
supportingResource	A ResourceRefOrValue [0..*]. A collection of resources that support this resource (bundling, link Resource Specification).
unknownStatus	A Boolean. The unknown status attribute (defined by ITU-T X.731/ITU-T M.3701) indicate that the state of the resource represented by the managed object is unknown. It is single valued and read-only by management functions.
usageState	A ResourceUsageStateType. Value enumeration for the 'resourceUsageState' of a Resource; the usage state attribute (defined by ITU-T X.731/ITU-T M.3701) tracks whether the resource is currently in use and if it can accept additional users or not; it is single-valued and read-only by management functions; defined values: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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Feature sub-resource fields

Configuration feature.

See [Feature] for details.

Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

See [Attachment] for details.

AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

See [AttachmentRef] for details.

Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : String Characteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, Number Characteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

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See [BooleanCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

See [IntegerArrayCharacteristic] for details.

IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s). If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the Characteristic.valueType property.

See [ObjectArrayCharacteristic] for details.

valueType sub-resource fields

Characteristic value.

See [valueType] for details.

ObjectCharacteristic sub-resource fields

A characteristic which value is an Object. If this characteristic is not specified by a CharacteristicSpecification and CharacteristicValueSpecification, and the value is not an Entity having an '@type' property, the type of the object MUST be specified by the

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Characteristic.valueType property.

See [ObjectCharacteristic] for details.

valueType_2 sub-resource fields

Value of the characteristic.

See [valueType_2] for details.

StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

A characteristic which value is a String.

See [StringCharacteristic] for details.

FeatureRelationship sub-resource fields

Configuration feature.

See [FeatureRelationship] for details.

Note sub-resource fields

Extra information about a given entity.

See [Note] for details.

PolicyRef sub-resource fields

Reference to managed Policy object.

See [PolicyRef] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

A reference to a Party or PartyRole playing a specific role in context of the entity to which it is related.

See [RelatedPartyRefOrPartyRoleRef] for details.

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PartyRef sub-resource fields

A Party reference.

See [PartyRef] for details.

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

See [PartyRoleRef] for details.

CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

See [CalendarPeriod] for details.

HourPeriod sub-resource fields

Hour interval.

See [HourPeriod] for details.

ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

See [ExternalIdentifier] for details.

IntentRef sub-resource fields

Intent reference, for when Intent is used by other entities.

See [IntentRef] for details.

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RelatedPlaceRef sub-resource fields

Use role to distinguish the context of the place being referenced (such as `delivery-entrance`). Use `@referredType` to distinguish the specific Place sub-type (such as a `GeographicAddress`). Use `@type` to extend (eg: as RefOrValue).

See [RelatedPlaceRef] for details.

PlaceRef sub-resource fields

A reference to a physical Place.

See [PlaceRef] for details.

RelatedResourceOrderItem sub-resource fields

RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.

See [RelatedResourceOrderItem] for details.

ResourceRelationship sub-resource fields

Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

See [ResourceRelationship] for details.

ResourceRef sub-resource fields

Resource reference, for when Resource is used by other entities.

See [ResourceRef] for details.

ResourceSpecificationRef sub-resource fields

Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an

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abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

See [ResourceSpecificationRef] for details.

Resource sub-resource fields

Resource is an abstract entity that describes the common set of attributes shared by all concrete resources (e.g. TPE, EQUIPMENT) in the inventory.

See [Resource] for details.

PhysicalResource sub-resource fields

Physical resource is a type of resource that describes the common set of attributes shared by all concrete physical resources (e.g. EQUIPMENT) in the inventory.

See [PhysicalResource] for details.

Json representation sample(s)

Example LogicalResource for a subscriber MSISDN including characteristics and relationships.


```
{
  "@type": "LogicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
  "id": "8044",
  "category": "Mobile",
  "description": "A mobile subscriber number (MSISDN) assigned to a customer.
Represents the logical resource used to identify a subscriber in the GSM/LTE network.",
  "name": "MSISDN +44-7700-900123",
  "startOperatingDate": "2023-01-15T00:00:00Z",
  "endOperatingDate": "2026-01-15T00:00:00Z",
  "administrativeState": "unlocked",
  "operationalState": "enabled",
  "usageState": "active",
  "validFor": {
    "startDateTime": "2023-01-15T00:00:00Z",
    "endDateTime": "2026-01-15T00:00:00Z"
  },
  "resourceVersion": "1.0.0",
  "note": [
    {
      "@type": "Note",
      "id": "note-001",
      "author": "network-provisioning-system",
      "date": "2023-01-15T08:30:00Z",
```


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```
      "text": "MSISDN provisioned and activated on LTE network. Ported from previous
operator."
    }
  ],
  "resourceOrderItem": [
    {
      "@type": "RelatedResourceOrderItem",
      "@referredType": "ResourceOrderItem",
      "resourceOrderHref": "/tmf-api/resourceOrderManagement/v5/resourceOrder/ORD-2023-
0042",
      "resourceOrderId": "ORD-2023-0042",
      "itemAction": "add",
      "itemId": "1"
    }
  ],
  "place": [
    {
      "@type": "RelatedPlaceRef",
      "role": "RegisteredAddress",
      "place": {
        "id": "addr-9912",
        "href": "/tmf-api/geographicAddressManagement/v5/geographicAddress/addr-9912",
        "@type": "PlaceRef",
        "@referredType": "GeographicAddress",
        "name": "10 Downing Street, London, SW1A 2AA, UK"
      }
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "user",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "href": "/tmf-api/partyManagement/v5/partyRole/456",
        "id": "456",
        "name": "Jane Smith",
        "@referredType": "Employee"
      }
    }
  ],
  "supportingResource": [
    {
      "@type": "ResourceRef",
      "id": "545",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/545",
      "name": "SIM Card ICCID-8944110012345678901",
      "@referredType": "PhysicalResource"
    }
  ],
  "resourceRelationship": [
    {
      "@type": "ResourceRelationship",
      "relationshipType": "reliesOn",
      "resource": {
        "@type": "ResourceRef",
        "id": "34",
        "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/34",
        "name": "LTE Base Station Node-B-23",
        "@referredType": "PhysicalResource"
```


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```
      },
      "resourceRelationshipCharacteristic": [
        {
          "@type": "NumberCharacteristic",
          "@baseType": "Characteristic",
          "id": "char-223",
          "name": "signalStrength",
          "valueType": "number",
          "value": -75
        }
      ]
    }
  ],
  "resourceCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-001",
      "name": "msisdn",
      "valueType": "string",
      "value": "+447700900123"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-002",
      "name": "networkType",
      "valueType": "string",
      "value": "LTE"
    },
    {
      "@type": "NumberCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-003",
      "name": "portingDays",
      "valueType": "number",
      "value": 3
    }
  ],
  "attachment": [
    {
      "@type": "AttachmentRef",
      "href": "/tmf-api/documentManagement/v5/attachment/att-22",
      "id": "att-22",
      "name": "Service Activation Certificate",
      "@referredType": "Attachment"
    }
  ],
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-101",
    "id": "spec-101",
    "name": "MSISDN",
    "@referredType": "ResourceSpecification",
    "version": "2.1.0"
  },
  "activationFeature": [
    {
      "@type": "Feature",
      "id": "feat-001",
```


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```
      "name": "VoLTE",
      "isBundle": false,
      "isEnabled": true,
      "featureCharacteristic": [
        {
          "@type": "StringCharacteristic",
          "@baseType": "Characteristic",
          "id": "char-feat-001",
          "name": "codecProfile",
          "valueType": "string",
          "value": "AMR-WB"
        }
      ],
      "featureRelationship": [
        {
          "@type": "FeatureRelationship",
          "id": "feat-rel-001",
          "name": "requires",
          "relationshipType": "requires",
          "validFor": {
            "startDateTime": "2023-01-15T00:00:00Z",
            "endDateTime": "2026-01-15T00:00:00Z"
          }
        }
      ],
      "policyConstraint": [
        {
          "@type": "PolicyRef",
          "href": "/tmf-api/policyManagement/v5/policy/pol-012",
          "id": "pol-012",
          "name": "VoLTE QoS Policy",
          "@referredType": "Policy",
          "version": "1.0.0"
        }
      ]
    }
  ],
  "intent": {
    "@type": "IntentRef",
    "href": "/tmf-api/intentManagement/v5/intent/int-124",
    "id": "int-124",
    "name": "Ensure 99.9% network availability",
    "@referredType": "Intent"
  },
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "ext-001",
      "externalIdentifierType": "IMSI",
      "owner": "HomeNetworkOperator"
    }
  ]
}
```


Notification Resource Models

16 notifications are defined for this API.

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Notifications related to Resource:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

Notifications related to Monitor:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

Notifications related to PhysicalResource:

- Resource Create Event
- Resource Delete Event
- Resource Attribute Value Change Event
- Resource State Change Event

Notifications related to LogicalResource:

- Resource Create Event
- Resource Delete Event
- Resource Attribute Value Change Event
- Resource State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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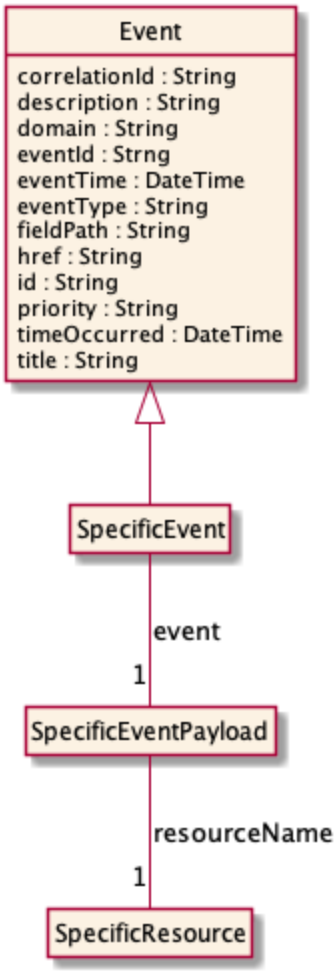


Figure 8 Notification Pattern

Resource

Create Event

Sample: ResourceCreateEvent for a PhysicalResource

Example notification when a PhysicalResource is created

```
Content-Type: application/json
{
  "@type": "ResourceCreateEvent",
  "eventId": "evt-res-create-001",
  "id": "evt-res-create-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceCreateEvent/evt-res-create-
001",
  "correlationId": "corr-2026-0331-001",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Created",
  "description": "A new PhysicalResource (Nokia 7750 SR-12 router) has been provisioned
```

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```
at Leeds data centre.",
  "priority": "1",
  "eventTime": "2026-03-31T10:00:00Z",
  "timeOccurred": "2026-03-31T10:00:00Z",
  "eventType": "ResourceCreateEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001",
      "id": "pr-4001",
      "name": "SR12-LDSC-01",
      "category": "NetworkEquipment",
      "description": "Nokia 7750 SR-12 core router at Leeds data centre.",
      "administrativeState": "locked",
      "operationalState": "disabled",
      "usageState": "idle",
      "resourceVersion": "1.0.0",
      "serialNumber": "NK2026L7SR12",
      "resourceSpecification": {
        "@type": "ResourceSpecificationRef",
        "id": "spec-204",
        "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-204",
        "name": "Nokia 7750 SR-12",
        "@referredType": "ResourceSpecification",
        "version": "1.0.0"
      }
    }
  }
}
```


Sample: ResourceCreateEvent for a LogicalResource

Example notification when a LogicalResource is created


```
Content-Type: application/json
{
  "@type": "ResourceCreateEvent",
  "eventId": "evt-res-create-002",
  "id": "evt-res-create-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceCreateEvent/evt-res-create-
002",
  "correlationId": "corr-2026-0331-002",
  "domain": "ResourceActivation",
  "title": "LogicalResource Created",
  "description": "A new LogicalResource (MSISDN) has been provisioned and activated on
the LTE network.",
  "priority": "1",
  "eventTime": "2026-03-31T11:15:00Z",
  "timeOccurred": "2026-03-31T11:15:00Z",
  "eventType": "ResourceCreateEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
```


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```
      "id": "8044",
      "name": "MSISDN +44-7700-900123",
      "category": "Mobile",
      "description": "A mobile subscriber number (MSISDN) provisioned and activated on
the LTE network.",
      "administrativeState": "unlocked",
      "operationalState": "enabled",
      "usageState": "active",
      "resourceVersion": "1.0.0",
      "resourceSpecification": {
        "@type": "ResourceSpecificationRef",
        "id": "spec-101",
        "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-101",
        "name": "MSISDN",
        "@referredType": "ResourceSpecification",
        "version": "2.1.0"
      }
    }
  }
}
```

Delete Event

Sample: ResourceDeleteEvent for a PhysicalResource

Example notification when a PhysicalResource is deleted

```
Content-Type: application/json
{
  "@type": "ResourceDeleteEvent",
  "eventId": "evt-res-del-001",
  "id": "evt-res-del-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceDeleteEvent/evt-res-del-
001",
  "correlationId": "corr-2026-0331-007",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Deleted",
  "description": "PhysicalResource CQURC has been decommissioned and removed from
inventory following end-of-life.",
  "priority": "1",
  "eventTime": "2026-03-31T16:00:00Z",
  "timeOccurred": "2026-03-31T16:00:00Z",
  "eventType": "ResourceDeleteEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/CQURC",
      "id": "CQURC"
    }
  }
}
```

Sample: ResourceDeleteEvent for a LogicalResource

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Example notification when a LogicalResource is deleted

```
Content-Type: application/json
{
  "@type": "ResourceDeleteEvent",
  "eventId": "evt-res-del-002",
  "id": "evt-res-del-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceDeleteEvent/evt-res-del-
002",
  "correlationId": "corr-2026-0331-008",
  "domain": "ResourceActivation",
  "title": "LogicalResource Deleted",
  "description": "LogicalResource (MSISDN) has been released and returned to the number
pool following contract termination.",
  "priority": "1",
  "eventTime": "2026-03-31T17:30:00Z",
  "timeOccurred": "2026-03-31T17:30:00Z",
  "eventType": "ResourceDeleteEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044"
    }
  }
}
```

Attribute Value Change Event

Sample: ResourceAttributeValueChangeEvent for a PhysicalResource

Example notification when a PhysicalResource attribute value changes

```
Content-Type: application/json
{
  "@type": "ResourceAttributeValueChangeEvent",
  "eventId": "evt-res-avc-001",
  "id": "evt-res-avc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceAttributeValueChangeEvent/ev
t-res-avc-001",
  "correlationId": "corr-2026-0331-003",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Attribute Value Changed",
  "description": "The name of PhysicalResource EEVWV has been updated to 'SR12-LDSC-01'
to align with the data centre naming convention.",
  "priority": "1",
  "eventTime": "2026-03-31T10:05:00Z",
  "timeOccurred": "2026-03-31T10:05:00Z",
  "eventType": "ResourceAttributeValueChangeEvent",
  "event": {
    "resource": {
```

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```
      "@type": "PhysicalResource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EEVWV",
      "id": "EEVWV",
      "name": "SR12-LDSC-01"
    }
  }
}
```

Sample: ResourceAttributeValueChangeEvent for a LogicalResource

Example notification when a LogicalResource attribute value changes

```
Content-Type: application/json
{
  "@type": "ResourceAttributeValueChangeEvent",
  "eventId": "evt-res-avc-002",
  "id": "evt-res-avc-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceAttributeValueChangeEvent/ev
t-res-avc-002",
  "correlationId": "corr-2026-0331-004",
  "domain": "ResourceActivation",
  "title": "LogicalResource Attribute Value Changed",
  "description": "The name of LogicalResource 8044 (an MSISDN allocation) has been
updated to 'MSISDN +44-7700-900123'.",
  "priority": "1",
  "eventTime": "2026-03-31T11:20:00Z",
  "timeOccurred": "2026-03-31T11:20:00Z",
  "eventType": "ResourceAttributeValueChangeEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044",
      "name": "MSISDN +44-7700-900123"
    }
  }
}
```

State Change Event

Sample: ResourceStateChangeEvent for a PhysicalResource

Example notification when a PhysicalResource state changes

```
Content-Type: application/json
{
  "@type": "ResourceStateChangeEvent",
  "eventId": "evt-res-sc-001",
  "id": "evt-res-sc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceStateChangeEvent/evt-res-sc-
```

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```
001",
  "correlationId": "corr-2026-0331-005",
  "domain": "ResourceActivation",
  "title": "PhysicalResource State Changed",
  "description": "The PhysicalResource ESPBW has been locked for scheduled
maintenance.",
  "priority": "2",
  "eventTime": "2026-03-31T22:00:00Z",
  "timeOccurred": "2026-03-31T22:00:00Z",
  "eventType": "ResourceStateChangeEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/ESPBW",
      "id": "ESPBW",
      "administrativeState": "locked"
    }
  }
}
```

Sample: ResourceStateChangeEvent for a LogicalResource

Example notification when a LogicalResource state changes

```
Content-Type: application/json
{
  "@type": "ResourceStateChangeEvent",
  "eventId": "evt-res-sc-002",
  "id": "evt-res-sc-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceStateChangeEvent/evt-res-sc-
002",
  "correlationId": "corr-2026-0331-006",
  "domain": "ResourceActivation",
  "title": "LogicalResource State Changed",
  "description": "The LogicalResource (MSISDN) has been suspended due to non-payment of
the associated service.",
  "priority": "1",
  "eventTime": "2026-03-31T13:45:00Z",
  "timeOccurred": "2026-03-31T13:45:00Z",
  "eventType": "ResourceStateChangeEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044",
      "administrativeState": "locked"
    }
  }
}
```

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Monitor

Create Event

Example notification when a Monitor is created

```
Content-Type: application/json
{
  "@type": "MonitorCreateEvent",
  "eventId": "evt-mon-create-001",
  "id": "evt-mon-create-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/monitorCreateEvent/evt-mon-create-
001",
  "correlationId": "corr-2026-0331-009",
  "domain": "ResourceActivation",
  "title": "Monitor Created",
  "description": "A Monitor resource has been created to track the asynchronous
execution of a resource provisioning request.",
  "priority": "1",
  "eventTime": "2026-03-31T10:00:00Z",
  "timeOccurred": "2026-03-31T10:00:00Z",
  "eventType": "MonitorCreateEvent",
  "event": {
    "monitor": {
      "@type": "Monitor",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/monitor/FMEFR",
      "id": "FMEFR",
      "state": "inProgress",
      "request": {
        "method": "POST",
        "to": "/tmf-api/resourceActivationAndConfiguration/v5/resource",
        "header": [
          {
            "name": "Content-Type",
            "value": "application/json"
          }
        ]
      }
    }
  }
}
```

Delete Event

Example notification when a Monitor is deleted

```
Content-Type: application/json
{
  "@type": "MonitorDeleteEvent",
  "eventId": "evt-mon-del-001",
  "id": "evt-mon-del-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/monitorDeleteEvent/evt-mon-del-001",
```

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```
  "correlationId": "corr-2026-0331-012",
  "domain": "ResourceActivation",
  "title": "Monitor Deleted",
  "description": "Monitor FMEFR has been deleted following the completion and cleanup
of the resource provisioning task.",
  "priority": "1",
  "eventTime": "2026-03-31T10:05:00Z",
  "timeOccurred": "2026-03-31T10:05:00Z",
  "eventType": "MonitorDeleteEvent",
  "event": {
    "monitor": {
      "@type": "Monitor",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/monitor/FMEFR",
      "id": "FMEFR"
    }
  }
}
```

Attribute Value Change Event

Example notification when a Monitor attribute value changes

```
Content-Type: application/json
{
  "@type": "MonitorAttributeValueChangeEvent",
  "eventId": "evt-mon-avc-001",
  "id": "evt-mon-avc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/monitorAttributeValueChangeEvent/evt
-mon-avc-001",
  "correlationId": "corr-2026-0331-010",
  "domain": "ResourceActivation",
  "title": "Monitor Attribute Value Changed",
  "description": "The state of Monitor FMEFR has changed from 'inProgress' to
'completed' following successful resource provisioning.",
  "priority": "1",
  "eventTime": "2026-03-31T10:02:00Z",
  "timeOccurred": "2026-03-31T10:02:00Z",
  "eventType": "MonitorAttributeValueChangeEvent",
  "event": {
    "monitor": {
      "@type": "Monitor",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/monitor/FMEFR",
      "id": "FMEFR",
      "state": "completed"
    }
  }
}
```

State Change Event

Example notification when a Monitor state changes

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```
Content-Type: application/json
{
  "@type": "MonitorStateChangeEvent",
  "eventId": "evt-mon-sc-001",
  "id": "evt-mon-sc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/monitorStateChangeEvent/evt-mon-sc-
001",
  "correlationId": "corr-2026-0331-011",
  "domain": "ResourceActivation",
  "title": "Monitor State Changed",
  "description": "Monitor mon-1002 state has changed to 'inError' due to a conflict in
the PATCH request for resource EEVWV.",
  "priority": "1",
  "eventTime": "2026-03-31T10:30:00Z",
  "timeOccurred": "2026-03-31T10:30:00Z",
  "eventType": "MonitorStateChangeEvent",
  "event": {
    "monitor": {
      "@type": "Monitor",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/monitor/mon-1002",
      "id": "mon-1002",
      "state": "inError"
    }
  }
}
```

PhysicalResource

Resource Create Event

Sample: ResourceCreateEvent for a PhysicalResource

Example notification when a PhysicalResource is created

```
Content-Type: application/json
{
  "@type": "ResourceCreateEvent",
  "eventId": "evt-res-create-001",
  "id": "evt-res-create-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceCreateEvent/evt-res-create-
001",
  "correlationId": "corr-2026-0331-001",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Created",
  "description": "A new PhysicalResource (Nokia 7750 SR-12 router) has been provisioned
at Leeds data centre.",
  "priority": "1",
  "eventTime": "2026-03-31T10:00:00Z",
  "timeOccurred": "2026-03-31T10:00:00Z",
  "eventType": "ResourceCreateEvent",
  "event": {
    "resource": {
```

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```
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001",
      "id": "pr-4001",
      "name": "SR12-LDSC-01",
      "category": "NetworkEquipment",
      "description": "Nokia 7750 SR-12 core router at Leeds data centre.",
      "administrativeState": "locked",
      "operationalState": "disabled",
      "usageState": "idle",
      "resourceVersion": "1.0.0",
      "serialNumber": "NK2026L7SR12",
      "resourceSpecification": {
        "@type": "ResourceSpecificationRef",
        "id": "spec-204",
        "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-204",
        "name": "Nokia 7750 SR-12",
        "@referredType": "ResourceSpecification",
        "version": "1.0.0"
      }
    }
  }
}
```

Sample: ResourceCreateEvent for a LogicalResource

Example notification when a LogicalResource is created

```
Content-Type: application/json
{
  "@type": "ResourceCreateEvent",
  "eventId": "evt-res-create-002",
  "id": "evt-res-create-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceCreateEvent/evt-res-create-
002",
  "correlationId": "corr-2026-0331-002",
  "domain": "ResourceActivation",
  "title": "LogicalResource Created",
  "description": "A new LogicalResource (MSISDN) has been provisioned and activated on
the LTE network.",
  "priority": "1",
  "eventTime": "2026-03-31T11:15:00Z",
  "timeOccurred": "2026-03-31T11:15:00Z",
  "eventType": "ResourceCreateEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044",
      "name": "MSISDN +44-7700-900123",
      "category": "Mobile",
      "description": "A mobile subscriber number (MSISDN) provisioned and activated on
the LTE network.",
      "administrativeState": "unlocked",
      "operationalState": "enabled",
```

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```
      "usageState": "active",
      "resourceVersion": "1.0.0",
      "resourceSpecification": {
        "@type": "ResourceSpecificationRef",
        "id": "spec-101",
        "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-101",
        "name": "MSISDN",
        "@referredType": "ResourceSpecification",
        "version": "2.1.0"
      }
    }
  }
}
```

Resource Delete Event

Sample: ResourceDeleteEvent for a PhysicalResource

Example notification when a PhysicalResource is deleted

```
Content-Type: application/json
{
  "@type": "ResourceDeleteEvent",
  "eventId": "evt-res-del-001",
  "id": "evt-res-del-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceDeleteEvent/evt-res-del-
001",
  "correlationId": "corr-2026-0331-007",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Deleted",
  "description": "PhysicalResource CQURC has been decommissioned and removed from
inventory following end-of-life.",
  "priority": "1",
  "eventTime": "2026-03-31T16:00:00Z",
  "timeOccurred": "2026-03-31T16:00:00Z",
  "eventType": "ResourceDeleteEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/CQURC",
      "id": "CQURC"
    }
  }
}
```

Sample: ResourceDeleteEvent for a LogicalResource

Example notification when a LogicalResource is deleted

```
Content-Type: application/json
{
  "@type": "ResourceDeleteEvent",
```

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```
  "eventId": "evt-res-del-002",
  "id": "evt-res-del-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceDeleteEvent/evt-res-del-
002",
  "correlationId": "corr-2026-0331-008",
  "domain": "ResourceActivation",
  "title": "LogicalResource Deleted",
  "description": "LogicalResource (MSISDN) has been released and returned to the number
pool following contract termination.",
  "priority": "1",
  "eventTime": "2026-03-31T17:30:00Z",
  "timeOccurred": "2026-03-31T17:30:00Z",
  "eventType": "ResourceDeleteEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044"
    }
  }
}
```

Resource Attribute Value Change Event

Sample: ResourceAttributeValueChangeEvent for a PhysicalResource

Example notification when a PhysicalResource attribute value changes

```
Content-Type: application/json
{
  "@type": "ResourceAttributeValueChangeEvent",
  "eventId": "evt-res-avc-001",
  "id": "evt-res-avc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceAttributeValueChangeEvent/ev
t-res-avc-001",
  "correlationId": "corr-2026-0331-003",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Attribute Value Changed",
  "description": "The name of PhysicalResource EEVWV has been updated to 'SR12-LDSC-01'
to align with the data centre naming convention.",
  "priority": "1",
  "eventTime": "2026-03-31T10:05:00Z",
  "timeOccurred": "2026-03-31T10:05:00Z",
  "eventType": "ResourceAttributeValueChangeEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EEVWV",
      "id": "EEVWV",
      "name": "SR12-LDSC-01"
    }
  }
```

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```
}
```

Sample: ResourceAttributeValueChangeEvent for a LogicalResource

Example notification when a LogicalResource attribute value changes

```
Content-Type: application/json
{
  "@type": "ResourceAttributeValueChangeEvent",
  "eventId": "evt-res-avc-002",
  "id": "evt-res-avc-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceAttributeValueChangeEvent/ev
t-res-avc-002",
  "correlationId": "corr-2026-0331-004",
  "domain": "ResourceActivation",
  "title": "LogicalResource Attribute Value Changed",
  "description": "The name of LogicalResource 8044 (an MSISDN allocation) has been
updated to 'MSISDN +44-7700-900123'.",
  "priority": "1",
  "eventTime": "2026-03-31T11:20:00Z",
  "timeOccurred": "2026-03-31T11:20:00Z",
  "eventType": "ResourceAttributeValueChangeEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044",
      "name": "MSISDN +44-7700-900123"
    }
  }
}
```

Resource State Change Event

Sample: ResourceStateChangeEvent for a PhysicalResource

Example notification when a PhysicalResource state changes

```
Content-Type: application/json
{
  "@type": "ResourceStateChangeEvent",
  "eventId": "evt-res-sc-001",
  "id": "evt-res-sc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceStateChangeEvent/evt-res-sc-
001",
  "correlationId": "corr-2026-0331-005",
  "domain": "ResourceActivation",
  "title": "PhysicalResource State Changed",
  "description": "The PhysicalResource ESPBW has been locked for scheduled
maintenance.",
```

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```
  "priority": "2",
  "eventTime": "2026-03-31T22:00:00Z",
  "timeOccurred": "2026-03-31T22:00:00Z",
  "eventType": "ResourceStateChangeEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/ESPBW",
      "id": "ESPBW",
      "administrativeState": "locked"
    }
  }
}
```

Sample: ResourceStateChangeEvent for a LogicalResource

Example notification when a LogicalResource state changes

```
Content-Type: application/json
{
  "@type": "ResourceStateChangeEvent",
  "eventId": "evt-res-sc-002",
  "id": "evt-res-sc-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceStateChangeEvent/evt-res-sc-
002",
  "correlationId": "corr-2026-0331-006",
  "domain": "ResourceActivation",
  "title": "LogicalResource State Changed",
  "description": "The LogicalResource (MSISDN) has been suspended due to non-payment of
the associated service.",
  "priority": "1",
  "eventTime": "2026-03-31T13:45:00Z",
  "timeOccurred": "2026-03-31T13:45:00Z",
  "eventType": "ResourceStateChangeEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044",
      "administrativeState": "locked"
    }
  }
}
```

LogicalResource

Resource Create Event

Sample: ResourceCreateEvent for a PhysicalResource

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Example notification when a PhysicalResource is created

```
Content-Type: application/json
{
  "@type": "ResourceCreateEvent",
  "eventId": "evt-res-create-001",
  "id": "evt-res-create-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceCreateEvent/evt-res-create-
001",
  "correlationId": "corr-2026-0331-001",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Created",
  "description": "A new PhysicalResource (Nokia 7750 SR-12 router) has been provisioned
at Leeds data centre.",
  "priority": "1",
  "eventTime": "2026-03-31T10:00:00Z",
  "timeOccurred": "2026-03-31T10:00:00Z",
  "eventType": "ResourceCreateEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001",
      "id": "pr-4001",
      "name": "SR12-LDSC-01",
      "category": "NetworkEquipment",
      "description": "Nokia 7750 SR-12 core router at Leeds data centre.",
      "administrativeState": "locked",
      "operationalState": "disabled",
      "usageState": "idle",
      "resourceVersion": "1.0.0",
      "serialNumber": "NK2026L7SR12",
      "resourceSpecification": {
        "@type": "ResourceSpecificationRef",
        "id": "spec-204",
        "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-204",
        "name": "Nokia 7750 SR-12",
        "@referredType": "ResourceSpecification",
        "version": "1.0.0"
      }
    }
  }
}
```

Sample: ResourceCreateEvent for a LogicalResource

Example notification when a LogicalResource is created

```
Content-Type: application/json
{
  "@type": "ResourceCreateEvent",
  "eventId": "evt-res-create-002",
  "id": "evt-res-create-002",
  "href": "/tmf-
```

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```
api/resourceActivationAndConfiguration/v5/listener/resourceCreateEvent/evt-res-create-
002",
  "correlationId": "corr-2026-0331-002",
  "domain": "ResourceActivation",
  "title": "LogicalResource Created",
  "description": "A new LogicalResource (MSISDN) has been provisioned and activated on
the LTE network.",
  "priority": "1",
  "eventTime": "2026-03-31T11:15:00Z",
  "timeOccurred": "2026-03-31T11:15:00Z",
  "eventType": "ResourceCreateEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044",
      "name": "MSISDN +44-7700-900123",
      "category": "Mobile",
      "description": "A mobile subscriber number (MSISDN) provisioned and activated on
the LTE network.",
      "administrativeState": "unlocked",
      "operationalState": "enabled",
      "usageState": "active",
      "resourceVersion": "1.0.0",
      "resourceSpecification": {
        "@type": "ResourceSpecificationRef",
        "id": "spec-101",
        "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-101",
        "name": "MSISDN",
        "@referredType": "ResourceSpecification",
        "version": "2.1.0"
      }
    }
  }
}
```

Resource Delete Event

Sample: ResourceDeleteEvent for a PhysicalResource

Example notification when a PhysicalResource is deleted

```
Content-Type: application/json
{
  "@type": "ResourceDeleteEvent",
  "eventId": "evt-res-del-001",
  "id": "evt-res-del-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceDeleteEvent/evt-res-del-
001",
  "correlationId": "corr-2026-0331-007",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Deleted",
  "description": "PhysicalResource CQURC has been decommissioned and removed from
inventory following end-of-life.",
```

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```
  "priority": "1",
  "eventTime": "2026-03-31T16:00:00Z",
  "timeOccurred": "2026-03-31T16:00:00Z",
  "eventType": "ResourceDeleteEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/CQURC",
      "id": "CQURC"
    }
  }
}
```

Sample: ResourceDeleteEvent for a LogicalResource

Example notification when a LogicalResource is deleted

```
Content-Type: application/json
{
  "@type": "ResourceDeleteEvent",
  "eventId": "evt-res-del-002",
  "id": "evt-res-del-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceDeleteEvent/evt-res-del-
002",
  "correlationId": "corr-2026-0331-008",
  "domain": "ResourceActivation",
  "title": "LogicalResource Deleted",
  "description": "LogicalResource (MSISDN) has been released and returned to the number
pool following contract termination.",
  "priority": "1",
  "eventTime": "2026-03-31T17:30:00Z",
  "timeOccurred": "2026-03-31T17:30:00Z",
  "eventType": "ResourceDeleteEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044"
    }
  }
}
```

Resource Attribute Value Change Event

Sample: ResourceAttributeValueChangeEvent for a PhysicalResource

Example notification when a PhysicalResource attribute value changes

```
Content-Type: application/json
{
  "@type": "ResourceAttributeValueChangeEvent",
```

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```
  "eventId": "evt-res-avc-001",
  "id": "evt-res-avc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceAttributeValueChangeEvent/ev
t-res-avc-001",
  "correlationId": "corr-2026-0331-003",
  "domain": "ResourceActivation",
  "title": "PhysicalResource Attribute Value Changed",
  "description": "The name of PhysicalResource EEVWV has been updated to 'SR12-LDSC-01'
to align with the data centre naming convention.",
  "priority": "1",
  "eventTime": "2026-03-31T10:05:00Z",
  "timeOccurred": "2026-03-31T10:05:00Z",
  "eventType": "ResourceAttributeValueChangeEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EEVWV",
      "id": "EEVWV",
      "name": "SR12-LDSC-01"
    }
  }
}
```


Sample: ResourceAttributeValueChangeEvent for a LogicalResource

Example notification when a LogicalResource attribute value changes


```
Content-Type: application/json
{
  "@type": "ResourceAttributeValueChangeEvent",
  "eventId": "evt-res-avc-002",
  "id": "evt-res-avc-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceAttributeValueChangeEvent/ev
t-res-avc-002",
  "correlationId": "corr-2026-0331-004",
  "domain": "ResourceActivation",
  "title": "LogicalResource Attribute Value Changed",
  "description": "The name of LogicalResource 8044 (an MSISDN allocation) has been
updated to 'MSISDN +44-7700-900123'.",
  "priority": "1",
  "eventTime": "2026-03-31T11:20:00Z",
  "timeOccurred": "2026-03-31T11:20:00Z",
  "eventType": "ResourceAttributeValueChangeEvent",
  "event": {
    "resource": {
      "@type": "LogicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
      "id": "8044",
      "name": "MSISDN +44-7700-900123"
    }
  }
}
```


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Resource State Change Event

Sample: ResourceStateChangeEvent for a PhysicalResource

Example notification when a PhysicalResource state changes

```
Content-Type: application/json
{
  "@type": "ResourceStateChangeEvent",
  "eventId": "evt-res-sc-001",
  "id": "evt-res-sc-001",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceStateChangeEvent/evt-res-sc-
001",
  "correlationId": "corr-2026-0331-005",
  "domain": "ResourceActivation",
  "title": "PhysicalResource State Changed",
  "description": "The PhysicalResource ESPBW has been locked for scheduled
maintenance.",
  "priority": "2",
  "eventTime": "2026-03-31T22:00:00Z",
  "timeOccurred": "2026-03-31T22:00:00Z",
  "eventType": "ResourceStateChangeEvent",
  "event": {
    "resource": {
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/ESPBW",
      "id": "ESPBW",
      "administrativeState": "locked"
    }
  }
}
```

Sample: ResourceStateChangeEvent for a LogicalResource

Example notification when a LogicalResource state changes

```
Content-Type: application/json
{
  "@type": "ResourceStateChangeEvent",
  "eventId": "evt-res-sc-002",
  "id": "evt-res-sc-002",
  "href": "/tmf-
api/resourceActivationAndConfiguration/v5/listener/resourceStateChangeEvent/evt-res-sc-
002",
  "correlationId": "corr-2026-0331-006",
  "domain": "ResourceActivation",
  "title": "LogicalResource State Changed",
  "description": "The LogicalResource (MSISDN) has been suspended due to non-payment of
the associated service.",
  "priority": "1",
  "eventTime": "2026-03-31T13:45:00Z",
  "timeOccurred": "2026-03-31T13:45:00Z",
```

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```
   "eventType": "ResourceStateChangeEvent",
   "event": {
     "resource": {
       "@type": "LogicalResource",
       "@baseType": "Resource",
       "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/8044",
       "id": "8044",
       "administrativeState": "locked"
     }
   }
 }
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on Resource

Retrieves a resource by ID

`GET /resource/{id}?fields=…`

Description

This operation retrieves a Resource entity. Attribute selection is enabled for all first level attributes.

Usage samples

Example showing retrieval of a single PhysicalResource by its ID

Request

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```
GET /resource/EYYXI
Content-Type: application/json
```

Response

```
200

{
  "@type": "PhysicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EYYXI",
  "id": "EYYXI",
  "category": "NetworkEquipment",
  "description": "Cisco ASR 9001 core router installed at the Manchester data centre.
Provides IP/MPLS routing for the northern regional network.",
  "name": "ASR9001-MCDC-01",
  "startOperatingDate": "2022-03-01T00:00:00Z",
  "endOperatingDate": "2027-03-01T00:00:00Z",
  "administrativeState": "unlocked",
  "operationalState": "enabled",
  "usageState": "active",
  "resourceVersion": "1.0.0",
  "serialNumber": "FCZ2201X3TK",
  "manufactureDate": "2021-09-15T00:00:00Z",
  "powerState": "on",
  "versionNumber": "IOS-XR 7.5.2",
  "note": [
    {
      "@type": "Note",
      "id": "note-eyyxi-001",
      "author": "network-operations",
      "date": "2022-03-01T09:00:00Z",
      "text": "Device commissioned and integrated into regional MPLS backbone."
    }
  ],
  "resourceCharacteristic": [
    {
      "@type": "NumberCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-eyyxi-001",
      "name": "portCount",
      "valueType": "number",
      "value": 48
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-eyyxi-002",
      "name": "rackUnit",
      "valueType": "string",
      "value": "U10"
    }
  ],
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-202",
```

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```
    "id": "spec-202",
    "name": "Cisco ASR 9001",
    "@referredType": "ResourceSpecification",
    "version": "1.0.0"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "networkEngineer",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "href": "/tmf-api/partyManagement/v5/partyRole/pr-501",
        "id": "pr-501",
        "name": "Alice Johnson",
        "@referredType": "Employee"
      }
    }
  ]
}
```

List or find resource objects

`GET /resource?fields=…&{filtering}`

Description

This operation list Resource entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Example showing retrieval of a collection of PhysicalResources

Request

```
GET /resource
Content-Type: application/json
```

Response

```
200

[
  {
    "@type": "PhysicalResource",
    "@baseType": "Resource",
    "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EYYXI",
    "id": "EYYXI",
    "category": "NetworkEquipment",
    "description": "Cisco ASR 9001 core router at Manchester data centre.",
```

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```
    "name": "ASR9001-MCDC-01",
    "startOperatingDate": "2022-03-01T00:00:00Z",
    "endOperatingDate": "2027-03-01T00:00:00Z",
    "administrativeState": "unlocked",
    "operationalState": "enabled",
    "usageState": "active",
    "resourceVersion": "1.0.0",
    "serialNumber": "FCZ2201X3TK",
    "resourceSpecification": {
      "@type": "ResourceSpecificationRef",
      "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-202",
      "id": "spec-202",
      "name": "Cisco ASR 9001",
      "@referredType": "ResourceSpecification",
      "version": "1.0.0"
    }
  },
  {
    "@type": "PhysicalResource",
    "@baseType": "Resource",
    "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-3305",
    "id": "pr-3305",
    "category": "NetworkEquipment",
    "description": "Juniper MX480 edge router at Birmingham data centre providing
peering and transit connectivity.",
    "name": "MX480-BHAM-01",
    "startOperatingDate": "2021-07-10T00:00:00Z",
    "endOperatingDate": "2026-07-10T00:00:00Z",
    "administrativeState": "unlocked",
    "operationalState": "enabled",
    "usageState": "active",
    "resourceVersion": "1.0.0",
    "serialNumber": "JN2202B9FL",
    "resourceSpecification": {
      "@type": "ResourceSpecificationRef",
      "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-203",
      "id": "spec-203",
      "name": "Juniper MX480",
      "@referredType": "ResourceSpecification",
      "version": "1.0.0"
    }
  }
]
```


Creates a resource

POST /resource?fields=…&{filtering}

Description

This operation creates a Resource entity.

Mandatory Attributes

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Mandatory Attributes	Rule
@type	

Usage samples

Example showing creation of a new PhysicalResource

Request

```
POST /resource
Content-Type: application/json

{
  "@type": "PhysicalResource",
  "@baseType": "Resource",
  "category": "NetworkEquipment",
  "description": "Nokia 7750 SR-12 core router to be installed at the Leeds data centre
for regional traffic aggregation.",
  "name": "SR12-LDSC-01",
  "startOperatingDate": "2026-04-01T00:00:00Z",
  "endOperatingDate": "2031-04-01T00:00:00Z",
  "administrativeState": "locked",
  "operationalState": "disabled",
  "usageState": "idle",
  "resourceVersion": "1.0.0",
  "serialNumber": "NK2026L7SR12",
  "manufactureDate": "2025-12-01T00:00:00Z",
  "powerState": "off",
  "versionNumber": "SR OS 22.10.R1",
  "resourceCharacteristic": [
    {
      "@type": "NumberCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-new-001",
      "name": "portCount",
      "valueType": "number",
      "value": 12
    }
  ],
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-204",
    "id": "spec-204",
    "name": "Nokia 7750 SR-12",
    "@referredType": "ResourceSpecification",
    "version": "1.0.0"
  }
}
```

Response

```
201
```

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```
{
  "@type": "PhysicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001",
  "id": "pr-4001",
  "category": "NetworkEquipment",
  "description": "Nokia 7750 SR-12 core router to be installed at the Leeds data centre
for regional traffic aggregation.",
  "name": "SR12-LDSC-01",
  "startOperatingDate": "2026-04-01T00:00:00Z",
  "endOperatingDate": "2031-04-01T00:00:00Z",
  "administrativeState": "locked",
  "operationalState": "disabled",
  "usageState": "idle",
  "resourceVersion": "1.0.0",
  "serialNumber": "NK2026L7SR12",
  "manufactureDate": "2025-12-01T00:00:00Z",
  "powerState": "off",
  "versionNumber": "SR OS 22.10.R1",
  "resourceCharacteristic": [
    {
      "@type": "NumberCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-new-001",
      "name": "portCount",
      "valueType": "number",
      "value": 12
    }
  ],
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-204",
    "id": "spec-204",
    "name": "Nokia 7750 SR-12",
    "@referredType": "ResourceSpecification",
    "version": "1.0.0"
  }
}
```


Updates partially a resource

`PATCH /resource/{id}?fields=…`

Description

This operation allows partial updates of a Resource entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

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Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
activationFeature	
administrativeState	
alarmStatus	
allocationStatus	
attachment	
availabilityStatus	
category	
controlStatus	
creationDate	
description	
endOperatingDate	
externalIdentifier	
intent	
lifecycleState	
name	
note	
operationalState	
place	
proceduralStatus	
relatedParty	
resourceCharacteristic	
resourceOrderItem	

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Patchable Attributes	Rule
resourceRelationship	
resourceSpecification	
resourceVersion	
startOperatingDate	
supportingResource	
unknownStatus	
usageState	
validFor	

Usage samples

Example showing a partial update of a PhysicalResource using application/merge-patch+json

Request

```
PATCH /resource/EEVWV
Content-Type: application/merge-patch+json

{
  "@type": "PhysicalResource",
  "administrativeState": "unlocked"
}
```

Response

```
200

{
  "@type": "PhysicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EEVWV",
  "id": "EEVWV",
  "category": "NetworkEquipment",
  "description": "Nokia 7750 SR-12 core router installed at Leeds data centre.",
  "name": "SR12-LDSC-01",
  "startOperatingDate": "2026-04-01T00:00:00Z",
  "endOperatingDate": "2031-04-01T00:00:00Z",
  "administrativeState": "unlocked",
  "operationalState": "enabled",
  "usageState": "active",
  "resourceVersion": "1.0.0",
  "serialNumber": "NK2026L7SR12",
  "manufactureDate": "2025-12-01T00:00:00Z",
```

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```
  "powerState": "on",
  "versionNumber": "SR OS 22.10.R1",
  "note": [
    {
      "@type": "Note",
      "id": "note-eevwv-001",
      "author": "network-operations",
      "date": "2026-03-31T10:00:00Z",
      "text": "Device commissioned and activated after successful site acceptance
test."
    }
  ],
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-204",
    "id": "spec-204",
    "name": "Nokia 7750 SR-12",
    "@referredType": "ResourceSpecification",
    "version": "1.0.0"
  }
}
```

Example showing a partial update of a PhysicalResource using application/json-patch-query+json

Request

```
PATCH /resource/ESPBW
Content-Type: application/json-patch-query+json

[
  {
    "op": "replace",
    "path": "/resourceCharacteristic/value?name=softwareVersion",
    "value": "IOS-XR 7.9.2"
  },
  {
    "op": "replace",
    "path": "/note/text?id=note-espbw-001",
    "value": "Device locked for scheduled maintenance window. Window extended to 08:00
UTC."
  }
]
```

Response

```
200

{
  "@type": "PhysicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/ESPBW",
  "id": "ESPBW",
  "category": "NetworkEquipment",
```

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```
  "description": "Cisco ASR 9001 core router at Edinburgh data centre.",
  "name": "ASR9001-EDDC-01",
  "startOperatingDate": "2021-05-20T00:00:00Z",
  "endOperatingDate": "2026-05-20T00:00:00Z",
  "administrativeState": "locked",
  "operationalState": "enabled",
  "usageState": "idle",
  "resourceVersion": "1.0.0",
  "serialNumber": "FCZ1901A2MP",
  "powerState": "on",
  "resourceCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-espbw-001",
      "name": "softwareVersion",
      "valueType": "string",
      "value": "IOS-XR 7.9.2"
    },
    {
      "@type": "StringCharacteristic",
      "@baseType": "Characteristic",
      "id": "char-espbw-002",
      "name": "managementIpAddress",
      "valueType": "string",
      "value": "10.20.30.41"
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "note-espbw-001",
      "author": "maintenance-system",
      "date": "2026-03-31T22:00:00Z",
      "text": "Device locked for scheduled maintenance window. Window extended to 08:00
UTC."
    },
    {
      "@type": "Note",
      "id": "note-espbw-002",
      "author": "noc-operator",
      "date": "2026-03-31T21:55:00Z",
      "text": "Traffic drained to ASR9001-EDDC-02 ahead of the maintenance window."
    }
  ],
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-202",
    "id": "spec-202",
    "name": "Cisco ASR 9001",
    "@referredType": "ResourceSpecification",
    "version": "1.0.0"
  }
}
```


Example showing a partial update of a PhysicalResource using application/json-patch+json

Request

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```
PATCH /resource/UBTGF
Content-Type: application/json-patch+json

[
  {
    "op": "replace",
    "path": "/usageState",
    "value": "busy"
  },
  {
    "op": "replace",
    "path": "/operationalState",
    "value": "enabled"
  }
]
```

Response

```
200

{
  "@type": "PhysicalResource",
  "@baseType": "Resource",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/UBTGF",
  "id": "UBTGF",
  "category": "NetworkEquipment",
  "description": "Juniper MX480 edge router at Bristol data centre handling internet
peering.",
  "name": "MX480-BRDC-01",
  "startOperatingDate": "2023-02-14T00:00:00Z",
  "endOperatingDate": "2028-02-14T00:00:00Z",
  "administrativeState": "unlocked",
  "operationalState": "enabled",
  "usageState": "busy",
  "resourceVersion": "1.0.0",
  "serialNumber": "JN2303B4KP",
  "powerState": "on",
  "resourceSpecification": {
    "@type": "ResourceSpecificationRef",
    "href": "/tmf-api/resourceCatalogManagement/v5/resourceSpecification/spec-203",
    "id": "spec-203",
    "name": "Juniper MX480",
    "@referredType": "ResourceSpecification",
    "version": "1.0.0"
  }
}
```

Deletes a resource

DELETE /resource/{id}?{filtering}

Description

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This operation deletes a Resource entity.

Usage samples

Example showing deletion of a Resource by its ID

Request

```
DELETE /resource/CQURC
Content-Type: application/json
```

Response

```
204
```

Operations on Monitor

Retrieves a monitor by ID

`GET /monitor/{id}?fields=…`

Description

This operation retrieves a Monitor entity. Attribute selection is enabled for all first level attributes.

Usage samples

Example showing retrieval of a single Monitor by its ID

Request

```
GET /monitor/FMEFR
Content-Type: application/json
```

Response

```
200

{
  "@type": "Monitor",
  "href": "/tmf-api/resourceActivationAndConfiguration/v5/monitor/FMEFR",
  "id": "FMEFR",
  "request": {
```

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```
    "method": "POST",
    "to": "/tmf-api/resourceActivationAndConfiguration/v5/resource",
    "body": {
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "category": "NetworkEquipment",
      "name": "SR12-LDSC-01",
      "administrativeState": "locked"
    },
    "header": [
      {
        "name": "Content-Type",
        "value": "application/json"
      },
      {
        "name": "Accept",
        "value": "application/json"
      }
    ]
  },
  "response": {
    "statusCode": "201",
    "body": {
      "@type": "PhysicalResource",
      "@baseType": "Resource",
      "id": "pr-4001",
      "href": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001",
      "name": "SR12-LDSC-01"
    },
    "header": [
      {
        "name": "Content-Type",
        "value": "application/json"
      },
      {
        "name": "Location",
        "value": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001"
      }
    ]
  },
  "state": "completed",
  "sourceHref": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001"
}
```


List or find monitor objects

`GET /monitor?fields=…&{filtering}`

Description

This operation list Monitor entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

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Example showing retrieval of a collection of Monitors

Request

```
GET /monitor
Content-Type: application/json
```

Response

```
200

[
  {
    "@type": "Monitor",
    "href": "/tmf-api/resourceActivationAndConfiguration/v5/monitor/FMEFR",
    "id": "FMEFR",
    "request": {
      "method": "POST",
      "to": "/tmf-api/resourceActivationAndConfiguration/v5/resource",
      "header": [
        {
          "name": "Content-Type",
          "value": "application/json"
        }
      ]
    },
    "response": {
      "statusCode": "201",
      "header": [
        {
          "name": "Location",
          "value": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001"
        },
        {
          "name": "Content-Type",
          "value": "application/json"
        }
      ]
    },
    "state": "completed",
    "sourceHref": "/tmf-api/resourceActivationAndConfiguration/v5/resource/pr-4001"
  },
  {
    "@type": "Monitor",
    "href": "/tmf-api/resourceActivationAndConfiguration/v5/monitor/mon-1002",
    "id": "mon-1002",
    "request": {
      "method": "PATCH",
      "to": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EEVWV",
      "header": [
        {
          "name": "Content-Type",
          "value": "application/merge-patch+json"
        }
      ]
```

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```
    },
    "response": {
      "statusCode": "200",
      "header": [
        {
          "name": "Content-Type",
          "value": "application/json"
        }
      ]
    },
    "state": "completed",
    "sourceHref": "/tmf-api/resourceActivationAndConfiguration/v5/resource/EEVWV"
  }
]
```


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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

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```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

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Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
5.0.0	2026	Rajneesh Kumar Jio Platforms Ltd.	Migration of TMF702 Resource Activation and Configuration API to v5.0.0 format. Aligned data model with TMF639 Resource Inventory v5. Added support for LogicalResource and PhysicalResource polymorphism. Updated all examples and schemas to conform to TMF630 API guidelines.

Contributors to Document

Version Number	Date	Modified by	Description
5.0.0	2026	Rajneesh Kumar Jio Platforms Ltd.	Initial v5 migration

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